Idiosyncratic Firm Risk, Cash Holdings and Lumpy Investment*

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Abstract

This paper explores the hypothesis that the increase in firm-level risk is the main driver of the long-run increase in firms' cash holdings over the last decades. I document a novel stylized fact: the impact of firm level uncertainty on cash holdings is stronger for firms that belong to sectors with greater lumpiness in investment rates. To rationalize this empirical finding I build a dynamic model in which financially constrained firms face uncertainty about future revenues and are subject to fixed adjustment costs of investment. In the model the degree of investment lumpiness is key to amplify the effects of volatility on cash holdings. The model is estimated on the Compustat data and used in a number of counterfactual experiments. I find that the increase in firm level volatility can explain almost 90% of the observed increase in cash holdings in the period 1980-2010. Furthermore, if fixed adjustment costs are absent, the increase in cash generated by higher uncertainty is reduced by half.

JEL Classification: E22, D22, E32, G31.
Keywords: Financing constraints, cash holdings, heterogeneous firms, lumpy investment, idiosyncratic shocks.

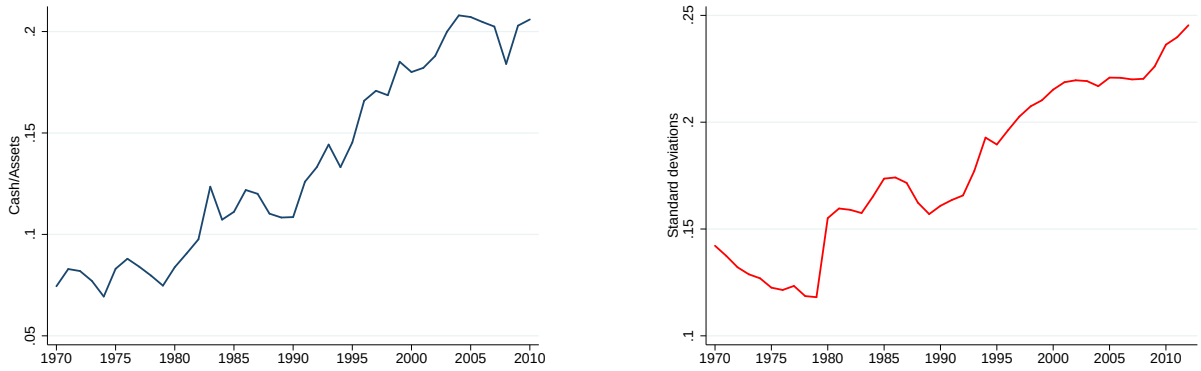
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1 Introduction

The U.S. corporate sector is holding record-high amounts of cash. Publicly traded U.S. firms have increased dramatically their cash holdings since the 1980s, except for a slowdown during the recent financial crisis, as it can be seen in Figure 1a. Over the time period 1980-2010, the share of cash holdings in the balance sheet of U.S. firms has increased from 10% to 22%. In 2011 these cash holdings amounted to nearly 5 trillion USD. A similar picture can be drawn for the evolution of net leverage at the firm level,[1] showing that net leverage (measured as total debt minus liquid assets) dropped from 17% into negative territory in the first decade of 2000s. These stylized facts on firms cash management challenge the traditional wisdom that views the corporate sector as a net borrower from the rest of the economy. Understanding this phenomenon can also shed some light on the reasons for the slow recovery from the Great Recession.

Figure 1: Stylized Facts, U.S. Economy



(a) Corporate Cash Holdings

(b) Firm Level Uncertainty

Notes: Panel (a) shows the evolution of the average cash holdings for U.S. public firms. Cash holdings are defined as cash and marketable securities (Compustat item CHE) divided by total assets (item `AT`). Panel (b) shows the volatility of cash flow growth at the firm level over 1970-2010. Cash flow is measured by operating income (Compustat item `OIBDP`). The volatility is computed as the standard deviation of cash flow growth over a ten-year rolling window and then averaged across firms. Data source: U.S. Compustat Annual, excluding financial companies and utilities.

Over the same time period, public firms experienced an upward trend in idiosyncratic uncertainty, as documented by Comin and Philippon (2005) and Davis et al. (2007). The increasing uncertainty in the firm's economic environment is reflected in various measures of firm volatility based on sales, cash flow, stock returns, etc. Figure 1b shows for example the volatility of the cash flow growth rate. This measure of cash flow volatility has almost

[1]For a more detailed analysis, see Section 2 of the paper.

doubled over the period 1980-2010, increasing from 0.15% to 0.25%. Other measures of firm-level risk deliver a similar message.[2]

Motivated by this evidence, in this paper I investigate the hypothesis that the increasing trend in firm volatility is the main driver of the secular increase in cash. The idea being analyzed is that financially constrained firms accumulate more cash in order to insure against negative profit shocks, and this mechanism can lead to significant accumulation of cash balances especially if firms perceive an increasing volatility in profits and cash flows. This precautionary motive for cash holding dates back to Keynes (1936)[3] and has been recently investigated by Han and Qiu (2007). It has been extended to a dynamic, quantitative setting in the important contribution of Riddick and Whited (2009).

At the same time, the effect of firm-level uncertainty on cash is vastly heterogeneous across firms and across sectors. In this paper I document a novel empirical finding: the positive effect of firm level uncertainty on cash holdings is stronger for firms that belong to sector with higher *investment lumpiness*.[4] As a preliminary step, I investigate several stylized facts that explain the link between the rise in firm level risk and the secular trend in corporate cash holdings. To this end I build on Comin and Philippon (2005) approach to construct a new set of firm level measures of uncertainty for all non-financial firms in Compustat between 1970 and 2010. I measure uncertainty at the firm level by looking at the cross-sectional dispersion of firm growth rate of sales, profit and TFP. These cross-sectional measured of uncertainty all exhibit an upward trend in the Compustat sample for the 1970-2010 time frame. Since such measures can be affected by compositional bias, I also build an alternative measure of firm level uncertainty based on computing the standard deviation in the time series with a five year rolling window. In the financial literature it is common to measure stock market volatility by computing the standard deviation of stock returns in a given time span. For example, Campbell et al. (2001) document an increase in the volatility of idiosyncratic stock returns. Using this alternative firm-level measure of uncertainty, I document several new empirical regularities on the link between firm financing and uncertainty.

In particular, there is a strong positive relation between firm risk and corporate cash both in the cross-section and in the time-series, with firm level shocks volatility emerging as the most important determinant of cash holdings. The key empirical fact that I

[2]See Section 2 for a more detailed analysis.

[3]Under the precautionary motive, corporations hold cash to hedge negative cash-flow shocks and avoid costly securities issuance.

[4]Investment behavior at the micro level is often lumpy: firms make a few large adjustments not very frequently. These large investment spikes account for most of the total investment activity.

subsequently discover is that the link between cash and firm level risk is especially strong for firms that are financially constrained and those that belong to sectors with greater investment inflexibility. Overall my empirical results suggest that firm level volatility has a pervasive impact on corporate cash holding decisions and that the impact is due to both financial and real frictions.

In order to better understand the economic forces that drive the empirical link between firm level volatility, cash holdings and investment lumpiness, I next develop a dynamic model of the joint determination of firm's financing, risk management, and capital accumulation decisions, extending the basic framework of Riddick and Whited (2009). The model is cast in an infinite horizon, discrete-time stochastic environment (without aggragate shocks) where firms make value maximizing investment decisions in real (capital) and financial (cash holdings) assets. There are two key frictions: first, financial frictions arise since equity financing involves issuance costs; second, there are real frictions since investment in capital is subject to non-convex adjustment costs, which make it infrequent and lumpy (see Caballero (1999) for a standard treatment).

In this setting I show that the interplay of external financing costs (in the form of costly equity issuances) and non-convex adjustment costs generates a quantitatively large precautionary demand for cash hoarding because it makes difficult for firms to generate funds by raising external finance when they have to pay a large fixed cost to invest. As a result, firms with positive investment opportunities tend to hold more cash in anticipation of Ss type adjustments in physical capital to avoid raising costly external finance. To assess the quantitative importance of this *precautionary saving channel*, I simulate the model and perform a counterfactual experiment by increasing the degree of volatility faced by firms: all else equal, a technological transformation that increases the turbolence of the economic environment to levels observed in the U.S. economy during the last decade can account for almost 90 percent of the increase in corporate cash holdings that are observed among public firms for the same period. Non-convex investment frictions prove to be crucial in generating this effect: a second counterfactual experiment where such real investment frictions are shut down shows that the increase in cash generated by higher uncertainty is reduced by almost half.

Related Literature. My paper is related to existing work along different lines of the macroeconomic and finance literature. First, it contributes to the recently growing literature on the so-called *cash puzzle* in the US economy. The role of uncertainty as a major determinant of corporate cash holdings was first analyzed in Opler et al. (1999)

and subsequentely by Bates et al. (2009) in various empirical settings. My works differs markedly from the above mentioned empirical studies since it investigates specifically the role of investment lumpiness in amplifying the precautionary motive for cash. Turning to more theoretical analysis, my modeling framework builds on Whited (2006), Riddick and Whited (2009) and DeAngelo et al. (2011). These works introduce corporate saving behavior into the standard Tobin's Q investment model,[5] showing that financing costs and uncertainty play a crucial role in determining the optimal saving policy of firms. However these papers do not consider the issue of rising corporate cash over time, focusing entirely on the cross-section. Moreover Whited (2006) and Riddick and Whited (2009) do not perform any structural estimation of their dynamic models, whereas I estimate model parameters using the simulated method of moments. Falato et al. (2013) propose the increase in asset intangibility as a plausible explanation behind the increase in cash holdings, whereas Begenau and Palazzo (2017) focus on R&D intensity. Regarding the relationship between uncertainty shocks and firm investment, the seminal contribution is Bloom (2009) who builds a heterogeneous firms model with non-convex adjustment costs, estimated to U.S. data. In his model, an increase in uncertainty leads to a drop in employment and output through a "freeze" in hiring and investment. A similar mechanism is explored by Bloom et al. (2018) in a much richer framework. My focus is however different since I investigate the relationship between uncertainty and cash (rather than investment) in the long-run, without considering business cycle frequencies.

My paper is also connected to an important strand of the literature in macroeconomics dealing with firm level volatility and its implications. Comin and Philippon (2005) and Comin and Mulani (2006) were the first to document the increasing trend in volatility among U.S. Compustat firms, from 1970s to 2000. Their finding was put into a broader perspective by Davis et al. (2007), who showed that private U.S. firms experienced instead a decrease in volatility over the same time period. The coexistence of increasing uncertainty among public firm and decreasing uncertainty among private firms has been further studied by Thesmar and Thoenig (2011). Whereas I do not focus explicitly on the ultimate reasons behind such long-run increase in uncertainty[6], I add to these studies by first extending the evidence on the increasing volatility of listed firms after 2000, and, second and more importantly, relating the uncertainty trend to the high levels of corporate cash.

[5]The first microfoundation of Tobin's Q investment model is due to Hayashi (1982). Hayashi's model was then extended to incorporate financial frictions by Gomes (2001).

[6]One prominent explanation scrutinized by Comin and Philippon (2005) is higher competition in the goods market. Building on this insight, Irvine and Pontiff (2009) document that deregulation in product markets has led to an increase in the idiosyncratic volatility of cash flows.

Outline of the paper. The paper proceeds as follows. In section 2 I document the empirical evidence regarding the relation between corporate cash holdings, firm-level risk and investment lumpiness. In section 3 I set up a structural model of heterogeneous firms with saving in cash and costly equity financing to rationalize the empirical findings detailed in the previous section. In section 4 I explain the estimation of the model and perform a number of counterfactual experiments, measuring the impact of the increase in firm-level uncertainty on cash holdings. Section 5 concludes the paper.

2 Empirical Evidence

This section summarizes a number of key stylized facts on the evolution of corporate cash holdings and firm level uncertainty over time. To this aim I assemble a large (unbalanced) panel of US firms over the 1970 to 2010 period. My sample includes all Compustat firm-year observations from 1970 to 2010 with positive values for (book) total assets and sales revenue for firms incorporated in the US. Financial firms (SIC code 6000-6999) and utilities (SIC codes 4900-4999) are excluded from the sample. This sample selection criteria are standard (see for example Bates et al. (2009)). In the end the sample comprises 19,065 firms for a total of 209,265 firm-year observations.

In the remainder of this section, I document the positive link between cash holdings and risk, and the fact that such impact is stronger for firms that belong to sectors with higher investment lumpiness and for firms that are more credit constrained.

Empirical Fact 1: *Firm cash holdings show a secular upward trend in 1970-2010, whereas net leverage (or net debt) show a decreasing trend over the same time span.*

It is widely recognized that US firms have been accumulating large piles of cash during the last three decades. Figure 2a plots the average ratio of cash to book assets over the sample period. It is evident from the figure that cash holdings display a pronounced upward trend increasing from 8 percent in 1970 to more than 20 percent in 2010. This evidence is consistent with the findings in Bates et al. (2009). There might be concerns that such rise in cash is driven only by a few but very large companies: however even if one considers the median level of cash ratio by year, or the aggregate cash to aggregate assets (which is average cash ratio weighted by assets[7]) the increasing trend is still there, as it is apparent from Figures 2a and 2b. More in detail, the fact that cash ratio weighted by firm size is lower than unweighted cash ratio suggests that larger firms hold on average

[7]Figure 2a (blue line) shows the evolutions of the average cash ratio, whereas Figure 2b shows instead the aggregate cash to aggregate asset ratio. For more details, see Appendix C.

less cash than small ones.

It is worth observing that such increasing trend in cash comes together with a decreasing trend in net leverage (see Figure 2c). While no clear trend is apparent if one looks at leverage, *net* leverage underwent a decrease from 19 percent in 1970 to negative values after 2004. For my purposes, considering *net* leverage as opposed to leverage is of fundamental importance for the following reason. Net leverage is computed at the firm level as (LT debt + current debt – Cash)/Book Assets, whereas leverage is computed as (LT debt + current debt)/Book Assets. It is more meaningful to consider net leverage as a measure of firm level financial leverage, and indeed by considering net leverage instead of leverage I get dramatically different conclusions. The net leverage series shows important business cycle fluctuations, but still the downward trend is apparent. Around 2004 net leverage becomes negative, which is a surprising fact given the conventional wisdom that considers the corporate sector as typically a net borrower.

I estimate a simple regression of the cash ratio on a constant and time to determine whether there exists a statistically significant trend in the cash ratio (see Figure 2d).

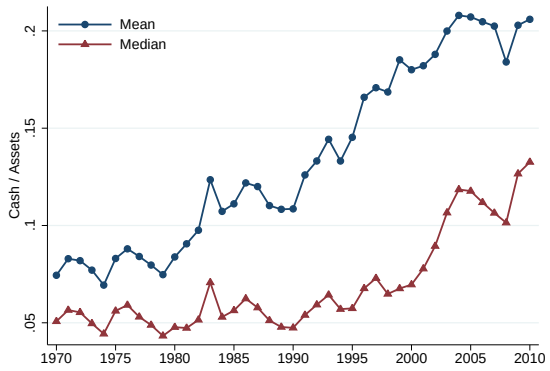
The coefficient for the time trend for the average cash ratio corresponds to a yearly increase of 0.39% and is highly statistically significant. The R^2 of the regression is 94%. For the median, the slope coefficient represents a 0.20% yearly increase. The R^2 is 71%. This evidence is consistent with a positive time trend in cash holdings during the sample period. When I regress instead net leverage (mean), which substracts cash from debt, on a constant and a time trend I get a decrease of 0.54%, again significative at the 99 percent level. Of course such simple regressions are only useful to characterize the evolution of the cash holdings during the sample period, and it would not make sense to extrapolate the in-sample trend to future years.

Empirical Fact 2: *The increasing trend in cash is more pronounced for* ***smaller*** *firms.*

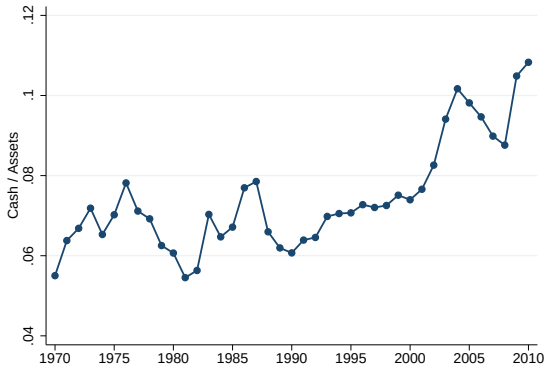
In Figure 3a I break the firms into quintiles of size (assets). Considering that small firms may find it harder to access credit markets, I would expect smaller firms to have higher cash-to-assets ratios. And indeed this intuition is confirmed in the data: smallest firms (those in the bottom quintiles of total assets, Q1/Q2) have on average higher cash-to-assets ratios. Moreover the increasing trend in cash ratios is more pronounced for smaller firms.

Empirical Fact 3: *The increasing trend in cash is more relevant for* ***non-paying dividend*** *firms.*

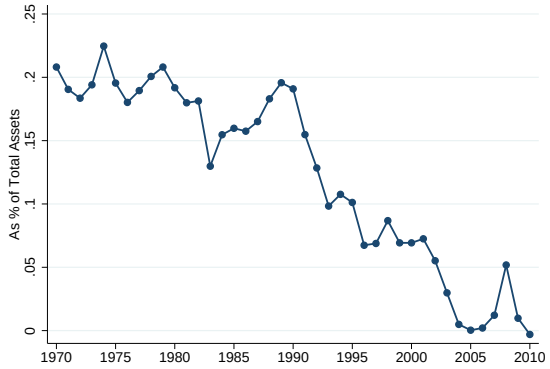
Figure 2: Rising Corporate Cash Holdings



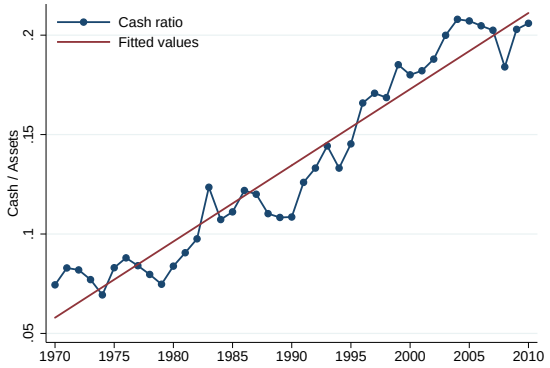
(a) Cash Holdings, Mean and Median



(b) Aggregate Cash



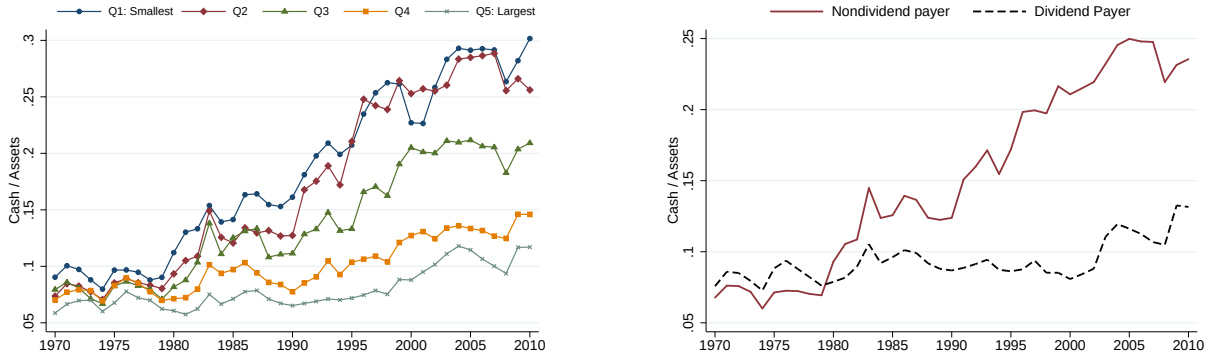
(c) Net Leverage



(d) Cash Holdings, Linear Trend

Notes: Cash holdings are defined for each firm as cash and marketable securities (Compustat item `CHE`) divided by total assets (item `AT`). Panel (a) shows the average and the median of firm-level cash holdings. Aggregate cash in panel (b) is defined for each year as total cash divided by total assets. Net leverage is computed as long-term debt (`DLTT`) + current debt (`DLC`) - cash divided by total assets. Data source: U.S. Compustat Annual, excluding financial companies and utilities.

Figure 3: Cash-to-assets ratio

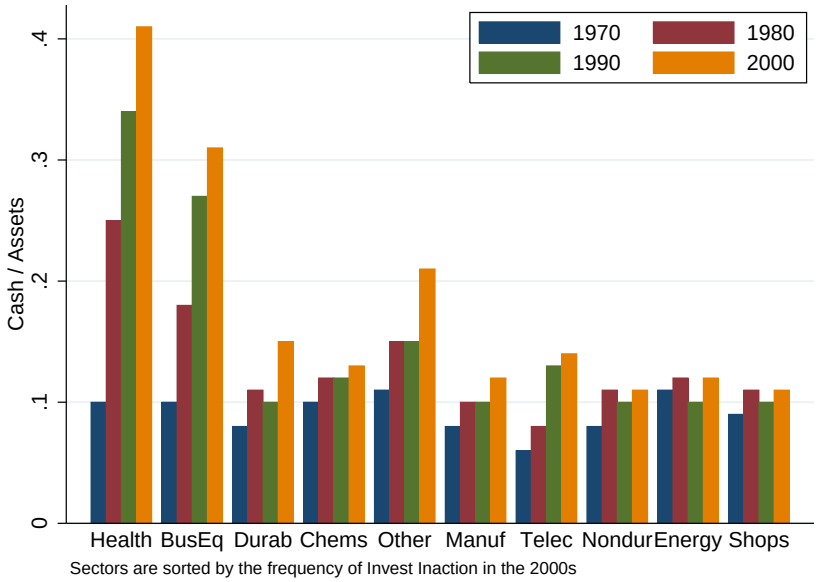


(a) by Firm Size

(b) by Dividend Paying Status

Notes: Firm size is measured as total assets. In panel (a), firms are split into quintiles of the size distribution. In panel (b) firms are split in two groups according to their dividend payer status. Dividend payer status is a dummy variable equal to one in years in which a firm pays a common dividend. Otherwise, the dummy equals zero. Data source: U.S. Compustat Annual, excluding financial companies and utilities.

Figure 4: Cash Holdings by Sectors



Notes: Sectors are sorted by the frequency of investment inaction in the 2000s and the vertical bars show average cash ratios for each decade by industry. Investment inaction is defined following Cooper and Haltiwanger (2006) as the number of firm-year observations such that $|\text{CAPX/AT}| < 1\%$ divided by the total number of observations in that sector. 2-digit sectors are grouped into 10 broad industries according to the Fama-French classification scheme. Data source: U.S. Compustat Annual, excluding financial companies and utilities.

Many papers in the corporate finance literature (for example, Almeida et al. (2004)) consider nondividend paying firms as firms that are more likely to be credit constrained. The evidence presented so far suggests that the increase in cash holdings occurred mostly in financially constrained firms.

From Figure 3b we can see that there is a dramatic increase in the cash ratio among the nondividend payers, but is much less so among the dividend payers. Under the aforementioned assumption that nondividend paying firms are more likely to be financially constrained, this evidence means that the increase in cash holdings occurred mostly in financially constrained firms.

Therefore if we consider *size* and *dividend payer status* as proxies for being financially constrained, empirical evidence supports the precautionary motive for holding cash.

Empirical Fact 4: *The increasing trend in cash is more relevant in sectors with higher investment lumpiness.*.

What about heterogeneity between sectors? While cash ratios have increased over time in all industries, the increase in cash was mostly driven by industrial sectors with high investment lumpiness.[8] These are sectors where the smooth adjustment of capital and labor is precluded by technological constraints such as fixed adjustment costs, (partial) irreversibility and indivisibilities. Figure 4 shows the evolution of average cash ratios by sectors. Sectors are sorted by the frequency of investment inaction[9] in 2000-2010 and the bars show average cash ratios for each decade by industry. For instance, capital adjustments are found to be smoother in "Shops" than in "Healthcare" or "Business Equipment". Cash ratios have increased over time in all industries but the most notable increases were observed in industries that had the highest investment lumpiness in the last decade. For example, cash ratios increased by 4 or 3 times in healthcare and business equipment but by much less in retail trade.

One might be worried that some other sector-specific factor (other than lumpiness) is responsible for the marked heterogeneity in the cash increase as shown in Figure 4. To address this concern, I provide more disaggregated evidence in Figure 11 in Appendix A. In this figure I split firms into the bottom and top quintiles of the frequency of investment inaction. Hence I assign to the high investment lumpiness group those firms in the top

[8]Lumpy investment behevior generates investment spikes, with little or no investment activity between the spikes. See for example Caballero et al. (1995).

[9]The frequency of investment inaction is a common proxy for investment lumpiness. The reason is that firms subject to higher fixed adjustment costs are more likely to change capital less frequently, leading to longer spells of inaction. Another proxy is the frequency of investment spikes: conditional on adjusting, firms adjust capital by a proportionally larger amount. See the Appendix for more details.

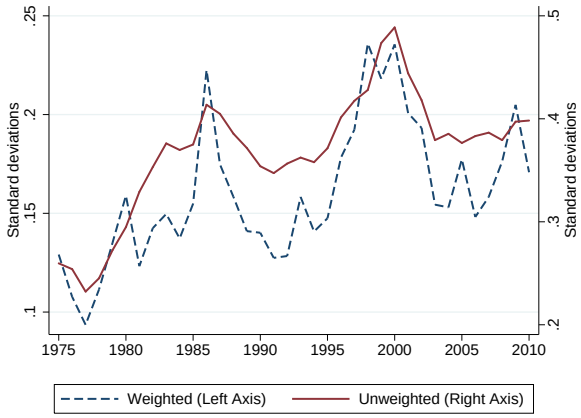
quintile of the frequency of investment inaction. The increase in cash is by far larger for firms in the top quintile, which increase cash-to-assets ratio by a factor of four.

Empirical Fact 4a: *Firm level risk showed an upward trend in last decades.*

As a first proxy to measure uncertainty at the firm level, I consider the cross-sectional dispersion of sales growth rate. The main finding is that the growth rate of sales has become more volatile over the sample period.

To measure the volatility inherent in the firm's environment I thus focus on the cross-section. Specifically I compute the standard deviations of growth rates across all the firms in a given year. As illustrated in Figure 5, volatility at the firm level exhibits a significant upward trend. In order to build a more representative measure of volatility, firm-level volatility measures are weighted using the firm's share of sales in total sales in a given year. Figure 5 also shows the persistent upward trend after the use of these weights.

Figure 5: Growth Rate of Firm-Level Sales: Cross-Sectional Dispersion



Notes: The figure shows the standard deviation of sales growth rates across all the firms in a given year, both weighted and unweighted. The weights are the firm's share of sales in total sales. Data source: Compustat Annual, excluding financial firms and utilities.

The source of this increase in volatility, however, may be subject to question. The upward trend may reflect a feature specific to sample in use, rather than reflecting a true change in the economy. Therefore I perform several robustness checks to validate this finding. First I adopt a different definition for the volatility of the growth rate of firm level sales. Instead of computing the dispersion of growth rate across all firms in a year (as previously shown in Figure 5), I define volatility as the time series of standard deviations of five-year rolling windows. In particular, for each firm i with more than five year of data I compute the centered 5-year rolling standard deviation of the firm-level growth rate of

sales:

$$\sigma_{it} = \left[\frac{1}{5} \sum_{\tau=-2}^{2} \left(\gamma_{i,t+\tau} - \overline{\gamma}_{it} \right)^2 \right]^{1/2},$$

where $\overline{\gamma}_{it}$ represents average sales growth between year $t-2$ and $t+2$ for firm i. I then take the median[10] across all firms present in the sample in a year to get an aggregate measure of the typical firm volatility in a given year:

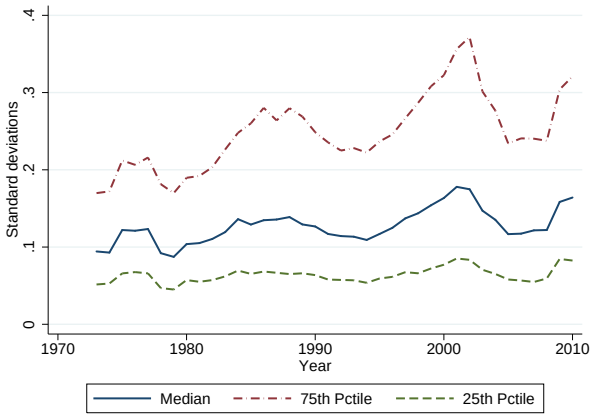
$$\sigma_t = median_i \left\{ \sigma_{it} \right\}.$$

This alternative definition of volatility is more suited to measure firm-level risk than the cross-sectional dispersion, and is less likely to be affected by compositional biases. When computing the standard deviation of the window in the time series, I remove the average growth rate for the firm in the window, and in effect control for firm-specific aspects that affect the growth rate of sales. These aspects, however, potentially show up in the cross-sectional measure. Figure 6b reports the finding using this second definition. As a robustness check I re-compute the rolling standard deviation of the sales growth rate using windows of shorter (3 year) and longer (10 year) lengths. Results are displayed in figures (6a) and (6c), respectively. The results are similar: an increasing trend in firm risk is evident, even though there seems to be a partial reversion in the trend after 2000.

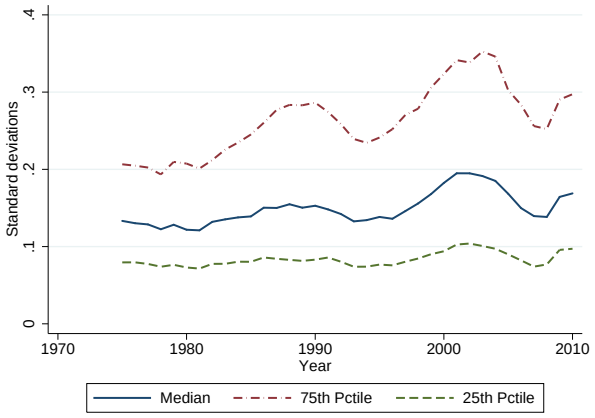
One might be concerned that this finding (i.e. the upward trend in volatility) is the result of changes in the composition of the Compustat sample over time. In this scenario, the upward trend in firm specific uncertainty could be simply the result of a larger share of smaller and younger firms, by construction more volatile, being listed in Compustat since the 1980s. I perform several robustness checks to confirm that is not the case. First, I restrict the sample to older firms that existed in the first ten years of my Compustat sample (from 1970 to 1980). This exercise follows loosely Acemoglu's discussion in Comin and Philippon (2005) NBER's chapter and its findings are depicted in Figure XXX in Appendix A. Second, I control for firm size and age, including firm fixed effects as well. The idea here is to isolate the component of volatility that is not explained by factors (such as size or age) that are changing in the sample. Results are shown in Figure 12 in Appendix A. All in all, these robustness checks support the finding that the trend in idiosyncratic sales volatility is still present, even though it is about 1/3 smaller.

[10] For comparison, Figure 1b in the Introduction depicts the average.

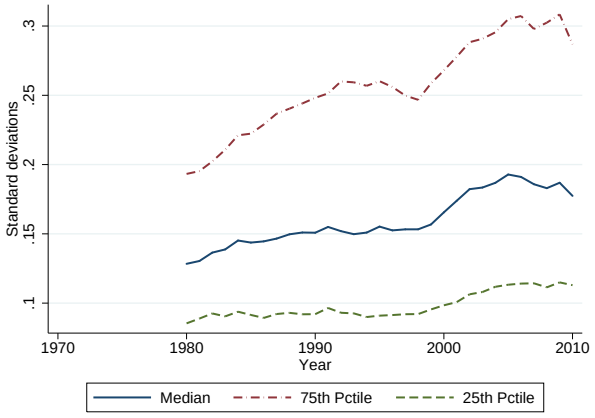
Figure 6: Time series Dispersion of the Growth rate of firm-level Sales



(a) 3-year rolling window



(b) 5-year rolling window



(c) 10-year rolling window

Notes: The panels shows the evolution of the median, 25th and 75th percentiles of firm volatility. Volatility is computed on a rolling window base with different widths (3,5 and 10 years). Data source: U.S. Compustat Annual, excluding financial companies and utilities.

2.1 Panel Evidence

In the remainder of this section, I confirm the stylized facts I presented above using panel data analysis. To this end I regress cash holdings on my preferred measure of firm-level uncertainty, while controlling for a set of standard determinants of cash holdings (see for example Bates et al. (2009) and Opler et al. (1999)). I consider several different variations of the following baseline specification:

$$Cash_{it} = \beta_0 + \beta_1 Risk_{it} + \beta_2' X_{it} + \delta_i + \lambda_t + \varepsilon_{it},$$

where $Cash_{it}$ is the ratio of firm i cash holdings to total assets in year t. The main explanatory variable is firm level uncertainty (measured as the rolling standard deviation of firm's growth rate of sales) and X_{it} is a vector of time-varying firm level controls that include market-to-book ratio, firm size, cash flow, capital expenditures, and a dummy for whether the firm pays dividend in a given year. I include year fixed-effects to control for time variation in cash holdings. The null hypothesis of my analysis is that β_1 equals zero. From Table 1 we can see that the coefficient on uncertainty is robustly positive and statistically significant in both OLS and FE specifications.

When I replicate the test for net leverage, which is the ratio of total debt net of cash to book assets, the coefficient on firm level risk is robustly negative and statistically significant across both specifications. These results suggest that firm level uncertainty is not only an important determinant of firms' cash holdings decisions, but also of their capital structure and net indebtness.

In the second part of this panel analysis (please refer to Tables 2 and 3) I use sample-split analysis to better understand why firm level volatility is an economically important determinant of corporate cash holdings. In particular, I examine both financial and real investment frictions, which are the key elements of my model (that I will explain in the next section).

Financial Frictions. If firms which operate in more volatile environments hold more cash for precautionary reasons, I would expect that the relation between uncertainty and cash is stronger among firms for which financing frictions are more severe.

Real Frictions. It is well know, since at least Bertola and Caballero (1994), that fixed adjustment costs lead firms to make large, lumpy investments. Therefore these real frictions may lead firms with higher uncertainty to accumulate even more cash to finance their large investments.

Indeed Table 2 shows evidence supporting the role of financial frictions. Here I follow the standard approach in the literature (see for example Whited and Wu (2006)) and in every year I rank firms based on two ex-ante indicators of their financial constraint status, which include firm size and dividend payer status. I assign to the financially contrained groups those firm in the bottom quartile of the annual distribution of each of these measures. Table 2 reports then the coefficient of firm level uncertainty on cash. Irrespective of which indicator of financing status is chosen, I find that the effect of uncertainty on cash is much stronger for firms that are more likely to face financial frictions.

In Table 3 instead I split the sample between bottom and top quartiles of the following two proxies of investment frictions: industry frequency of investment inaction and an indicator for whether there are investment spikes in the industry. For constructing these two indicators[11] I follow closely Cooper and Haltiwanger (2006). Consistently across the specifications and the type of indicator, the economic significance of the coefficient on uncertainty is much stronger for firms that are more likely to face real investment frictions.

In summary, the empirical regularities discovered by the panel analysis presented above are:

Empirical Fact 5: *Firm level risk is a major determinant of the increase in corporate cash holdings in the cross section.*

Empirical Fact 6: *The positive impact of firm level uncertainty on corporate cash holdings is higher for financially constrained firms and for firms belonging to sectors with higher lumpiness in investment.*

Why does a higher level of uncertainty lead firms to accumulate more cash? In this section I highlighted the role of financial and real frictions. As I summarized in the introduction, higher uncertainty increase the need to protect against negative shocks, but this effect is relevant as long as firm face credit constraints. Moreover, firms that are subject to large fixed costs when investing, are more likely to need external finance, since their cash flow from operations may not be enough to cover large capital expenditures. When the volatility of firm level shocks increases, firm become more uncertain about the timing of future investment opportunities, and anticipating the possibility of financial frictions, accumulate more cash and liquid assets.

[11] See Appendix C for more details.

Table 1: Panel evidence on firm level uncertainty and cash holdings

	Cash		**Net Leverage**	
	OLS	FE	OLS	FE
	(1)	(2)	(3)	(4)
Firm level risk σ_{it}^F	0.056***	0.048***	-0.122***	-0.039***
	(0.000)	(0.000)	(0.000)	(0.000)
Year fixed effects	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Adjusted R^2	0.299	0.672	0.325	0.601

Notes: The regression includes time dummies and standard controls for firm-level determinants of cash holdings: Tobin Q, firm size, cash flow, capital expenditures, acquisitions, dividend dummy. p-values are in parenthesis and are clustered at the firm level. Data source: U.S. Compustat Annual, excluding financial companies and utilities.

Table 2: Sample split analysis: Financial Frictions

	Panel A: Financial Frictions	
	OLS	FE
	(1)	(2)
By Firm Size		
Q1	0.088	0.036
	(0.009)**	(0.008)**
Q4	0.065	-0.001
	(0.010)**	(0.007)
By Dividend Payer Status		
No	0.114	0.026
	(0.007)**	(0.005)**
yes	0.041	0.001
	(0.009)**	(0.006)

Notes: The effect of uncertainty on cash holdings is stronger for financially constrained firms. The sample is split between bottom and top quartiles of the (yearly) distribution of: firm size and dividend payment status. The table reports point estimates of the coefficient of cash holdings on firm level uncertainty. The regression includes time dummies and standard controls for firm-level determinants of cash holdings. Standard errors are in parenthesis and are clustered at the firm level.

Table 3: Sample split analysis: Real Frictions

	Panel B: Real Frictions	
	OLS	FE
	(5)	(6)
By Frequency of Invest. Inaction		
Q1	0.075	0.011
	(0.007)**	(0.005)*
Q4	0.159	0.030
	(0.011)**	(0.009)**
By investment Spikes		
No	0.102	0.012
	(0.008)**	(0.006)*
yes	0.132	0.024
	(0.008)**	(0.005)**

Notes: The effect of uncertainty on cash holdings is stronger for firms with higher investment lumpiness. The sample is split between first and last quartile of (year): frequency of investment inaction - $|\text{Capex/Assets}| < 0.01$, and whether in the industry there are investment spikes - $|\text{Capex/Assets}| > 0.2$, based on Cooper and Haltiwanger (2006). The table reports point estimates of the coefficient of cash holdings on firm level uncertainty. The regression includes time dummies and standard controls for firm-level determinants of cash holdings. Standard errors are in parenthesis and are clustered at the firm level.

3 The Model

In this section I develop a dynamic model of the joint determination of firm's financing, cash management, and capital accumulation decisions, extending the framework of Gomes (2001) and Riddick and Whited (2009) along several dimensions. The aim is to use the model to assess the contribution of the increase in firm level risk observed in the 1980-2010 period on the increase in corporate cash holdings over the same time span. The economic mechanism (as explained in Section 1) relies on firms precautionary saving: firms accumulate cash as a buffer to finance future investment opportunities that come infrequently due to non-convex capital adjustment costs. The dynamic model contains therefore the following features:

- Firms with *heterogeneous* productivity levels.
- *Financial frictions*: External funds are more costly than internal funds. If firms could simple borrow at no additional cost whenever funds become necessary, they would not need to hold cash for precautionary reasons.
- *Real frictions*: Firms face non-convex costs of capital adjustment. Lumpy investment would reinforce the precautionary motive of holding cash. In the data the increase

in cash holdings is stronger for firms that are more likely to face real frictions. I ask then if the model is able to match also this related stylized fact.

3.1 Technology

I begin with describing the economic problem of the firms. Firms are ex-ante identical and are subject to an idiosyncratic productivity shock denoted by z_{it}.[12] I assume that these shocks are generated by a first-order Markov process with transition probability $Q(z', z)$, where a prime indicates next period variables. More specifically I assume an autoregressive process with persistence ρ:

$$\log z' = \rho \log z + \varepsilon', \tag{1}$$

where $\varepsilon' \sim N(0, \sigma_\varepsilon^2)$. As it is standard in the literature, I discretize the continuous time process described in (1) as a Markov chain with transition matrix Q using the method proposed by Tauchen (1986). The transition matrix Q is such that $\Pr\{z' = z_j | z = z_i\} = Q_{ij} \geq 0$ and $\sum_j Q_{ij} = 1$ for each $i = 1, \ldots, N_z$. Even though firms are ex-ante identical they differ ex-post since they experience different histories of idiosyncratic productivity shocks. Later in my empirical analysis, in order to make the actual data comparable with the simulated data generated from the model, I use firm and time fixed effects: the idea is to eliminate from the actual data all the heterogeneity that is not accounted for by the model.[13]

The firm's per period profits are denoted with $\pi(k, z)$. It is not restrictive to assume that the profit function is simply a function of (k, z) since it is understood that variable factors of production, such as labor, have already been maximized out. In other words,

$$\pi(k, z) = \max_n \{f(k, n, z) - wn\},$$

where $f(k, n, z)$ is the production function w is the wage rate and n is labor (or any other variable production factor). I assume the profit function is strictly increasing in each argument and strictly concave in capital, i.e. $\pi_k(k, z) > 0$, $\pi_z(k, z) > 0$ and $\pi_{kk}(k, z) < 0$. An example that satisfies these properties is given by $\pi(k, z) = zk^\alpha$, with $0 < \alpha < 1$. The empirical counterpart of $\pi(k, z)$ in Compustat is *operating cash flows* (`OIBDP`).

[12] The shocks z_{it} could in principle capture any shock affecting firm's revenues, hence not only shocks to technical efficiency but also idiosyncratic demand shocks. Unfortunately, the Compustat database does not contain firm-level prices.

[13] In fact the simulations of the model produce independent and identically distributed firms where the only heterogeneity comes from the realizations of the shocks. When matching the model to the data, it is therefore important to extract as much heterogeneity as possible.

Investment in capital is defined as:

$$I\left(k,k'\right)=k'-\left(1-\delta\right)k,$$

where $\delta\in(0,1)$ is the depreciation rate of the capital stock. When a firm buys or sells capital it also incurs adjustment costs:

$$\psi(I,k)=\frac{\psi_0}{2}\left(\frac{I}{k}\right)^2 k+F\cdot\mathbb{1}_{\{I\neq 0\}}k, \tag{2}$$

where the first term on the right-hand side of (2) denotes *convex* adjustment costs whereas the second term denotes *fixed* adjustment costs. The functional form in (2) is standard in the structural investment literature (see for example Cooper and Haltiwanger (2006)). Without convex adjustment costs the firm would react too quickly to productivity shocks. The presence of fixed costs capture any instances in which the firm bears costs that are independent on the size of the adjustment. It is needed to account for infrequent adjustment and the presence of many firm-year observations with zero investment in the real firm level data. The presence of fixed cost of adjusting capital is also crucial for explaining the cross-sectional dynamics of cash holdings over time, namely the empirical finding that firms belonging to sectors with more investment lumpiness accumulated more cash in response to an increase in volatility. Indeed the presence of fixed costs make the firm investment policy lumpy (i.e. the firm invests or disinvests only for large enough shocks, and stay inactive otherwise); large and infrequent investments typically require large debt or equity issuances. If these external financing is costly, then firm holds cash for precautionary reasons.

3.2 Financing

I now turn to the firm's financing and saving choices. Crucially capital is *not* the only way to transfer resources across time periods. The firm can save in the form of cash holdings $p\in\mathbb{R}_+$. Saving in cash is modeled as investing in a riskless one-period bond that earns taxable interest at a rate r. Firms are also subject to corporate taxation in the form of a flat tax τ_c. As in the U.S. tax code, both profits and interest on cash holdings are taxed at rate τ_c, but depreciation expenses are tax deductible. The firm taxable income can then be written as:

$$y=\pi\left(k,z\right)-\delta k+rp$$

and cash flow is defined as the firm's after-tax income $(1-\tau_c)\,y$ plus depreciation:

$$\text{CF} = (1-\tau_c)\,\pi\,(k,z) + \tau_c \delta k + r\,(1-\tau_c)\,p. \tag{3}$$

Cash flow together with the choices for capital and cash in the next period pin down the gross equity payout[14] distributed (if positive) to shareholders:

$$e = CF - [k' - (1-\delta)\,k] - \psi(I,k) - [p' - p]\,. \tag{4}$$

Firms can finance investment either with internal funds or borrowing from the financial market, by raising new equity. By raising external finance the firm incurs a variety of additional costs going from flotation costs to adverse selection premia. As in Gomes (2001) and Hennessy and Whited (2007) I do not model explicitly a setting with asymmetric information but I capture the simple fact that external funds are more costly than internal funds in a reduced form way. In particular, I assume that the additional cost of raising external finance is given by:

$$\lambda\,(e) = \begin{cases} \lambda_0 + \lambda_1\,|e| + \lambda_2 e^2, & \text{if } e < 0 \\ 0, & \text{if } e \geq 0. \end{cases} \tag{5}$$

The interpretation is the following: if $e \geq 0$ the firm is making distributions to shareholders, and if instead $e < 0$ the firm is issuing new equity. In this latter case the firm has to pay an additional cost given by a fixed cost λ_0, a per unit linear cost λ_1 and a per unit quadratic cost λ_2. A large body of empirical research provides detailed evidence regarding underwriting fees (see, among others, Altinkilic and Hansen (2000)) finding that there are significant economies of scale for small to medium issuances but then costs increase; therefore a fomulation with convex cost function that generates a U-shaped average cost seems most appropriate. In any case the simulated method of moments is used to estimate the three parameters of (5).

Finally the net equity payout is given by:

$$(1-\tau_d \mathbb{1}_{e\geq 0})e - \lambda(e).$$

[14] Substituting 3 into 4 shows that the gross equity payout can be rewritten as:

$$\begin{aligned} e = {} & (1-\tau_c)\,\pi\,(k,z) + \tau_c \delta k \\ & - [k' - (1-\delta)\,k] - \psi\,(I,k) \\ & - p' + [1 + r\,(1-\tau_c)\,p]\,. \end{aligned}$$

This captures the fact that if the gross equity payout e is positive then the firm pays out dividends, subject to the dividend tax

$$d = (1 - \tau_d)\, e.$$

If instead the gross equity payout is negative, the firm has to raise external finance and pays the extra borrowing cost associated to it:

$$\text{eqiss} = e - \left(\lambda_0 + \lambda_1 |e| + \lambda_2 e^2\right) < 0,$$

where eqiss denote equity issuances including the extra costs. Absent any form of financial market imperfections, cash would have no value for the firm; in this framework cash derives value because it is an alternative to costly external finance (see the subsequent analysis in Section 3.3 for more on this point).

The firm problem is to choose investment and financial policy to maximize net payments to its shareholders, taking as given the interest rate r (which is the opportunity cost of funds):

$$V(k, p, z) = \max_{k', p'} \left\{ \left(1 - \tau_d \mathbb{1}_{\{e>0\}}\right) e - \lambda(e) + \frac{1}{1+r} E_{z'|z} V(k', p', z') \right\} \tag{6}$$

s.t.

$$e = (1 - \tau_c)\, \pi(k, z) + \tau_c \delta k - I - \psi(I, k) - p' + (1 + r(1 - \tau_c))\, p, \tag{7}$$

$$k' = (1 - \delta)\, k + I, \tag{8}$$

$$\psi(I, k) = \frac{\psi_0}{2} \left(\frac{I}{k}\right)^2 k + F \cdot \mathbb{1}_{\{I \neq 0\}}, \tag{9}$$

$$\lambda(e) = \left(\lambda_0 + \lambda_1 |e| + \frac{1}{2}\lambda_2 e^2\right) \mathbb{1}_{\{e<0\}}, \tag{10}$$

$$0 \leq p' \leq \bar{p}. \tag{11}$$

The term $E_{z'|z} V(k', p', z')$ is the continuation value associated to a given choice of future capital and future cash and can be written more extensively as

$$E_{z'|z} V(k', p', z') = \int_{z'} V(k', p', z')\, dQ(z', z).$$

Notice that the upper bound in (11) makes the choice set for p compact but is never binding because of the tax penalty associated with holding cash. Problem (6) admits a

unique solution given the existence of upper bounds $\overline{p}$ and $\overline{k}$ for cash balances and capital.[15]

Equation (7) represents a sources and uses of funds identity: dividends/issuances are equal to cash flow from operations minus capital expenditures (the term $I(k,k')+\psi(I,k)$) minus net increases in cash (the term $p'-(1+r(1-\tau_c))p$). Recall that the gross equity payout e can be either positive or negative. If positive it represents distribution of internal cash flows to shareholders (i.e. dividends), and if negative it represents infusions of cash flows from shareholders into the firm. Hence if capital expenditures exceed internal cash flows the firm can either raise costly external finance (incurring the additional cost) or decrease the stock of cash holdings.

3.3 Economic mechanism

In this section I develop the economic intuition behind the model by examining its optimality conditions (although the model has no analytical form solution). The value function is in general not differentiable due to the presence of the cost of issuing equity, but for the sake of exposition I assume that V is concave and once differentiable.[16] See Appendix B for all derivations.

Denoting with μ the multiplier attached to the non-negativity constraint for p, the Lagrangian is:

$$\mathcal{L}=\left(1-\tau_d\mathbb{1}_{\{e>0\}}\right)e-\lambda(e)+\frac{1}{1+r}E_{z'|z}V(k',p',z')+\mu' p',$$

with

$$e(k,k',p,p',z)\equiv(1-\tau_c)\pi(k,z)+\tau_c\delta k-I(k',k)-\psi(k',k)-p'+(1+r(1-\tau_c))p$$

and

$$\begin{aligned}\lambda(e)&=\left(\lambda_0+\lambda_1|e|+\frac{1}{2}\lambda_2 e^2\right)\mathbb{1}_{\{e<0\}}\\&=\left(\lambda_0-\lambda_1 e+\frac{1}{2}\lambda_2 e^2\right)\mathbb{1}_{\{e<0\}}.\end{aligned}$$

[15] Indeed if we define the current return function as $F(k,k',p,p',z)=\left(1-\tau_d\mathbb{1}_{\{e(k,k',p,p',z)>0\}}\right)e-\lambda(e(k,k',p,p',z))$ then problem 6 satisfies the assumptions of Theorem 9.6 in Stokey, Lucas and Prescott (1989). In fact the existence of upper bounds for capital and cash and the assumption that the shock z is discrete ensure that F is defined over the compact set $\left[0,\overline{k}\right]^2\times[0,\overline{p}]^2\times\mathcal{Z}$, and therefore since it is continuous it is bounded.

[16] The optimality conditions for the choice of cash balances are obtained under the further assumptions that $\tau_d=0$ and $\lambda_0=0$.

The first order condition for p' is:

$$1 + (\lambda_1 - \lambda_2 e)\, \mathbb{1}_{\{e<0\}} = \frac{1}{1+r} E_{z'|z} \left[V_p \left(k', p', z' \right) \right] + \mu'. \tag{12}$$

The left-hand side of (12) are the costs of holding more cash balances into the next period, which include the equity issuance costs if the current period payoff is negative. The right-hand-side of (12) are instead the benefits related to additional cash.

The envelope condition reads as:

$$V_p \left(k, p, z \right) = \left(1 + r \left(1 - \tau_c \right) \right) \left[1 + (\lambda_1 - \lambda_2 e)\, \mathbb{1}_{\{e<0\}} \right].$$

The above condition summarizes the benefits of entering the current period with more cash holdings carried over from the last period. In particular the benefits take into account the tax penalty (i.e. interests on cash are taxed at rate τ_c) but also the savings on issuance costs if this period payoff e turns out to be negative.

Updating the envelope condition and plugging it into the first order condition (12) delivers:

$$1 + (\lambda_1 - \lambda_2 e)\, \mathbb{1}_{\{e<0\}} = \left(\frac{1 + r \left(1 - \tau_c \right)}{1+r} \right) E_{z'|z} \left[1 + (\lambda_1 - \lambda_2 e')\, \mathbb{1}_{\{e'<0\}} \right] + \mu'. \tag{13}$$

If external finance is costless (i.e. $\lambda_1 = \lambda_2$) then the optimality condition (13) becomes

$$1 - \frac{1 + r \left(1 - \tau_c \right)}{1+r} = \mu',$$

which implies $\mu' > 0$ and hence $p' = 0$ by complementary slackness. Hence the firm optimally holds no cash if external finance is free. However, in presence of credit frictions, holding cash increases the *financial flexibility* of the firm; by increasing saving by a marginal amount today, the firm entails the tax penalty but also reduces the probability of having to issue external finance tomorrow. The firm invests in cash up to the point where the marginal gain from reducing external finance costs equates the tax penalty on saving.

Figures 7a and 7b help visualize the optimal financial policy of the firm. Figure 7a shows the marginal cost and marginal benefit of a marginal increase in next-period cash balances, p', under the hypothesis that financing frictions are absent. In such a case the marginal benefit is given by the term $\frac{1+r(1-\tau_c)}{1+r} < 1$ (see also equation 13) whereas the marginal cost is simply 1. Trivially, since the marginal cost is always higher than the marginal benefit,

the optimal choice of p' and hence p will be zero. Figure 7b shows instead the more interesting case of costly external finance: now the marginal cost is a weakly increasing and piecewise differentiable curve whereas the marginal benefit is decreasing. It is useful to split the financing situation of the firm in three regions:[17]

- *Dividend distribution regime*: for low values of p' the firm is able to pay dividends to its shareholders and does not need to raise any extra equity. In this case the marginal cost of cash is 1 (or $1 - \tau_d$ if dividend taxes are positive).

- *Inaction regime*: for intermediate values of p' the firms retains all internal funds and does not distribute dividends nor issue new equity. In this case when $e = 0$ the marginal cost belongs to the interval $[1, 1 + \lambda_1]$ (or $[1 - \tau_d, 1 + \lambda_1]$ if dividend taxation is considered).

- *Equity issuance regime*: for high values of p' the firm is likely to run a financing deficit (i.e. $e < 0$) and has to raise additional equity. Since equity issuances are not free, the marginal cost is now higher and equal to $1 + \lambda_1 - \lambda_2 e$ (this is represented by the increasing line in Figure 7b).

The marginal benefit is represented instead as a decreasing curve in Figure 7b. In fact, from equation (13) it can be seen that the marginal benefit of cash is given by

$$\left(\frac{1 + r\,(1 - \tau_c)}{1 + r}\right) E_{z'|z}\left[1 + (\lambda_1 - \lambda_2 e')\,\mathbb{1}_{\{e' < 0\}}\right]$$

A higher level of next-period cash balances reduces the likelihood of incurring into a negative payoff in the next period (in other words, the probability of the event $\{e' < 0\}$ decreases) and having to tap costly external finance. Thus the financial flexibility benefit of cash is decreasing in p'.

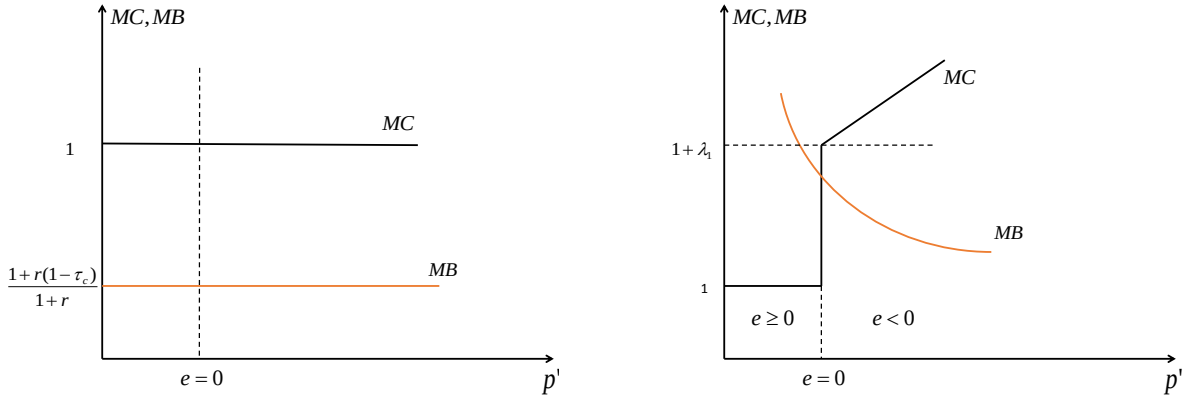
The optimal choice for an interior p' occurs at the intersection between the marginal cost and the marginal benefit schedule. Depending on the particular parametrization, a corner solution at $p' = 0$ may also occur.

To further inspect the economic incentives for investment and cash in the model, Figure 8 shows the policy functions for investment I, next-period cash balances p' and equity payout e along the profitability dimension (i.e. varying $\log(z)$, for given values of current capital k and cash p). Figure 9 depicts instead the optimal equity payout as a function of both capital and productivity, fixing only cash at a given value[18]. The message here is that

[17] See Appendix B for the formal derivations.

[18] Other policy functions are shown in the Appendix.

Figure 7: Cash holdings: optimal choice

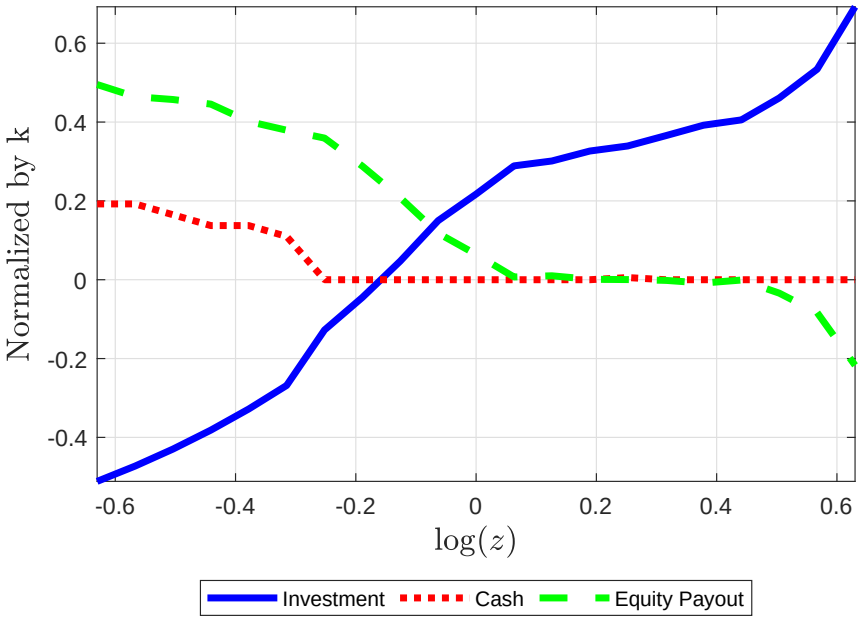


(a) No financial frictions

(b) Costly equity finance

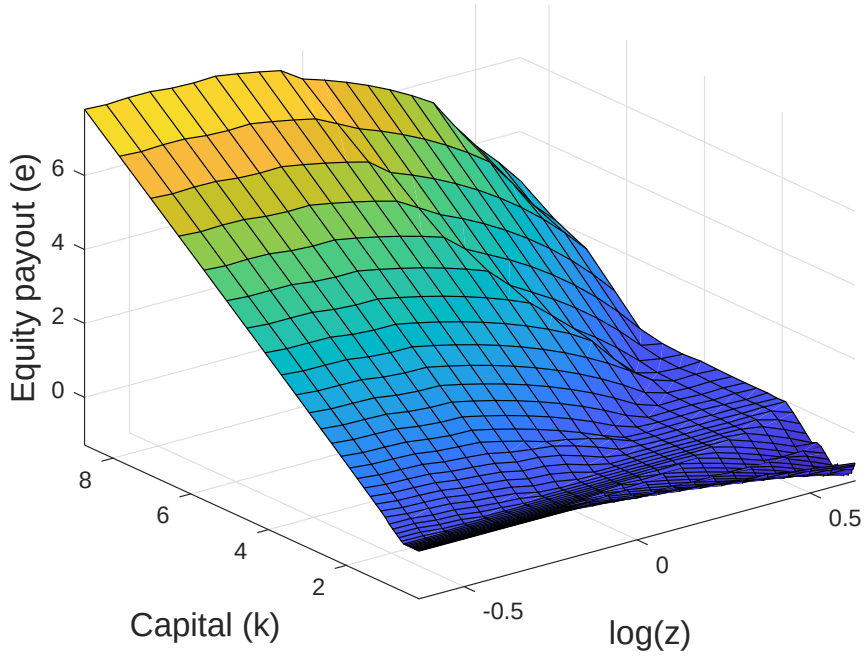
Notes: The figures depict the optimality conditions for the choice of next-period cash balances p'. See equation (13) in the main text. Panel (a) shows graphically equation (13) if equity issuances are costless, while panel (b) shows equation (13) if equity issuances are costly.

Figure 8: Policy Functions for Investment, Cash Holdings and Equity Payout



Notes: The figure shows the policy functions for investment, cash and equity payout as defined in the text. Each of the three variables is scaled by the capital stock. The policy functions are evaluated at the unconditional mean of capital and cash, along $\log(z)$, the profitability dimension.

Figure 9: Policy Function for Equity Payout



Notes: The figure shows the policy function for equity payout $e(k, p, z)$ as a function of capital k and productivity $\log(z)$. Equity payout is scaled by capital stock. The policy function is evaluated at the unconditional average of cash in the model simulations.

investment in physical capital and in cash are closely substitutes. As current productivity z rises, the firm optimally invests a greater deal of capital and brings down cash holdings to finance these investment opportunities. The firm chooses instead to increase the level of cash balances when the productivity of capital is relatively low.

Before presenting the calibration and the results in the next section, I delve more into the economics behind the model by exploring the simulated moments as a function of a subset of parameters. In particular I examine the sensitivity of average cash holdings, defined as $p/(k+p)$, to some model parameters (see Figure 10). The key intuition delivered from this exercise is that any model feature that raises the probability of needing external financing will also raise the precautionary motive of holding cash.

The top panels of Figure 10 illustrate the sensitivity of cash holdings to each of the external finance parameters.[19] It is evident that cash increases with the fixed and linear component of the external finance function λ_0 and λ_1, because the value of financial flexibility increases as external finance becomes more costly. When either λ_0 or λ_1 is zero then the optimal level of cash balances is also zero. Then as the cost parameters increase,

[19]However it seems that the effect of the quadratic component λ_2 (not reported in the figure) is much less clear-cut: the relation between average cash holdings and λ_2 is almost flat: with λ_0 and λ_1 set at their baseline levels, the effect of λ_2 is of second-order importance.

optimal cash holdings increase sharply but then flatten out. This means that financing constraints alone are not able to generate the sizable increase in cash observed in the data.

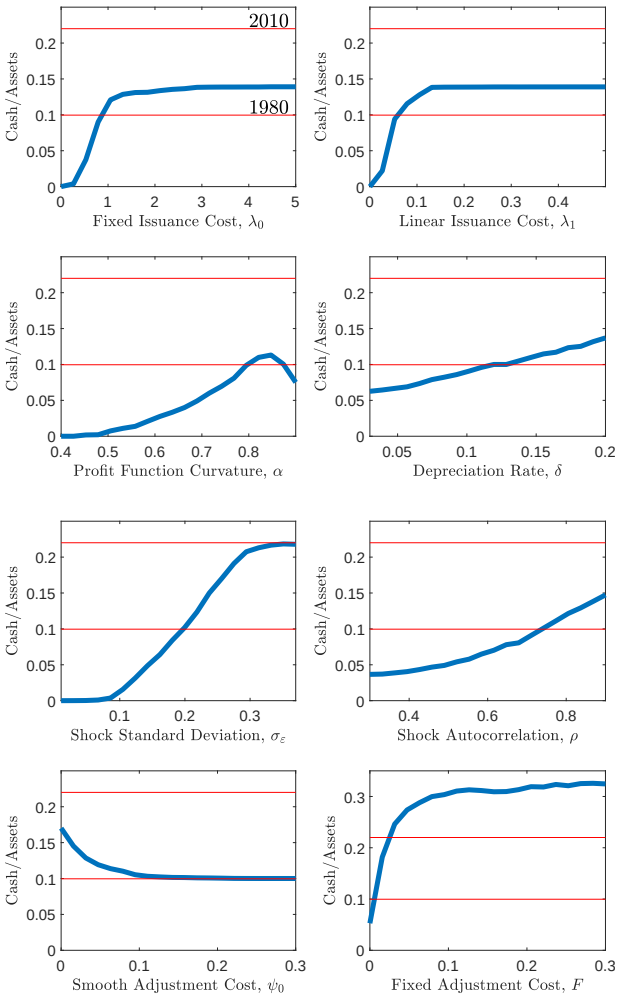
Turning our attention to the second row of the figure, we see how the profit function curvature (i.e. the capital coefficient α in the production function) and the depreciation rate δ affect cash balances. A higher depreciation rate monotonically increases cash holdings, whereas the relation between the capital coeffcient α and cash is hump-shaped, revealing the presence of two contrasting effects. First, as α rises, the average size of investment rises and hence the firm has to hold more cash to finance greater investment needs. Second, a higher profit curvature α implies that for given capital the firm generates larger internal revenues (i.e. the term $\pi(z,k)$ is larger) which reduces the likelihood of needing external finance, and hence the firm needs to hold less cash.

Focusing now on the third row of Figure 10, it is shown that cash holdings increase in both the variance σ_ε^2 and the serial correlation ρ_ε of the idiosyncratic shocks: firms facing more volatile or persistent shocks are more likely to need external funds for investment, and hence optimally accumulate more cash as a buffer.

Finally, the bottom panel of Figure 10 shows that smooth capital adjustment costs and fixed adjustment costs affect cash holdings in opposite directions[20]. With high convex adjustment costs the firm invests only to replace depreciated capital, never expects to need outside financing and therefore holds no cash, as it is shown in the left bottom panel. The right panel shows instead that larger fixed costs of adjusting capital induce optimally higher cash holdings. This effect occurs because higher fixed adjustment costs lead to larger investment that occur less frequently. The firm then uses episodes of inaction to accumulate cash, which acts to lower the probability of the firm having to raise external equity when it does invest.

[20]This feature turns out to be very useful in the identification of the adjustment cost parameters, when the model is calibrated to the firm level data.

Figure 10: The Cash/Assets ratio as a function of parameters



Notes: To construct each graph, I solve and simulate the model for each parameter value and then I compute the average ratio of cash across simulations. The two red lines denote the level of cash/assets observed in the Compustat data in 1980-1985 and 2005-2010, respectively.

4 Results

Since the model admits no closed form solution, I solve and simulate it with the help of numerical techniques. In order to find a numerical solution I need to resort to value function iteration on a discrete grid with interpolation. The numerical solution cannot be accomplished with either policy function iteration since the policy function is not continuous (due to the presence of fixed costs) or methods based on a low level polynomial approximation since the value function needs to be approximated well on a wide range of values of k. Therefore I implement the less efficient but robust method of value function iteration on a discrete state space. I specify a grid with a finite number of values for the capital stock k and the cash stock p, together with upper bounds $\overline{k}$ and $\overline{p}$. I choose the

upper bound for capital $\overline{k}$ to be such that

$$\pi\left(\overline{k}, z\right) - \delta\overline{k} = 0.$$

Because $k > \overline{k}$ is not economically profitable, k lies in the interval $(0, \overline{k}]$. The grid for capital is made of n_k points and is defined as follows:

$$K = \left\{\overline{k}, \overline{k}\left(1-\delta\right), \overline{k}\left(1-\delta\right)^2, \ldots, \overline{k}\left(1-\delta\right)^{n_k-1}\right\}.$$

Given the presence of fixed adjustment costs $F \cdot \mathbb{1}_{\{k'-(1-\delta)k\neq 0\}}k$, it is crucial that the capital grid includes the point $k' = k\left(1-\delta\right)$ for every k to account for investment inaction exactly.[21]

The upper bound for cash holdings is chosen so that $\overline{p} = \overline{k}$. The grid for cash holdings is made of $n_p = \frac{n_k}{2}$ points and it is equispaced. Recall that in the model without costly external finance the tax penalty of cash would induce the firm to choose $p = 0$. With financial frictions the firm engages in precautionary saving behaviour letting $p > 0$ but nonetheless the optimal choice for p never hits the upper bound $\overline{p}$ in the simulations.

The unknown parameters of the model are estimated using the simulated method of moments (SMM). This procedure selects model parameters in order to minimize the distance between the a given number of data moments and the corresponding model-generated moments. With the estimated version of the structural model at hand, I subsequently perform a counterfactual simulation feeding into the model the exogenous increase in firm-level uncertainty observed over the 1980-2010 period, but keeping everything else constant. In this way I can quantify the relative contribution of increasing uncertainty to the increasing cash holdings at the corporate level.

4.1 Data

The data used for the estimation are based on the Compustat sample described in Section 2. The sample is an unbalanced panel of U.S. firms covering from 1980 to 2010.[22] I exclude firm-year observations for which total assets, sales or capital are either negative or missing.

[21] In the computation the capital grid is actually constructed in a slightly different form as:

$$K = \left\{\overline{k}, \overline{k}(1-\delta)^{\chi}, \overline{k}(1-\delta)^{2\chi}, \ldots, \overline{k}(1-\delta)^{(n_k-1)\chi}\right\},$$

where $\chi > 0$. The computational parameter χ governs the width of the grid: if $\chi > 1$ then the grid is streched out, if $\chi < 1$ then the grid becomes more concentrated.

[22] The data for the 1970-1980 decade are only used for illustrative purposes to generate Figure 1a and Figures XXX in Section 2.

I also drop observations with total assets less than 10 million USD. Finally I delete firms belonging to SIC codes 6000-6999, 4900-4999 or greater than 9000 since my model is ill-suited to study regulated, financial firms or utilities. I trim the top and bottom 2% of the variables following Hennessy and Whited (2007). After the data cleaning criteria are applied, there are a total of 133803 firm-year observations with 14256 unique firms.

Data variables are defined as follows: book assets is Compustat item `AT`; capital stock is item `PPENT`; investment is `CAPX`; equity issuance is defined as equity issuances net of repurchases (`SSTK-PRSTKC`) and cash flow is defined as income before extraordinary items plus depreciation (`IB+DP` in Compustat).

The cross-sectional statistics are determined by considering each firm-year as a data point. See Appendix C for more details on the variables definition both in the Compustat data and in the model.

4.2 Selection of Targets and Identification

The model is estimated over a cross-section of firms comprising 5 years, from 1980 to 1984. I parametrize the model at an annual frequency using the SMM as estimation method, with the exception of three parameters that are instead taken from other studies or directly from the data. Regarding these parameters set outside the model, I fix the interest rate r at 4% by taking the average of the real interest rate[23] over the period 1980-1984. The dividend tax rate τ_d and the corporate tax rate τ_c are instead taken from Hennessy and Whited (2007). These choices are summarized in Table 4.

Table 4: External Parameters

Parameter	Description	Value
r	Real interest rate	0.04
τ_c	Corporate tax rate	0.46
τ_d	Dividend tax rate	0.12

After fixing the parameters in Table 4, the remaining nine parameters need to be estimated inside the model:

$$\theta = \{\alpha,\ \delta,\ \sigma_\varepsilon,\ \rho,\ \psi_0,\ F,\ \lambda_0,\ \lambda_1,\ \lambda_2\}\,. \tag{14}$$

These are the production function curvature α, the depreciation rate of capital δ, the

[23]The real interest rate in the data is computed as the difference between the 3 months interest rate on government bonds and the rate of inflation.

standard deviation and the autocorrelation of the productivity shock $\{\sigma_\varepsilon, \rho\}$, the adjustment cost parameters $\{\psi, F\}$ and the parameters related to the external finance costs $\{\lambda_0, \lambda_1, \lambda_2\}$. The model is overidentified in the sense that these 9 parameters are chosen to match 13 data targets (discussed below). The simulated method of moments procedure works as follows. Given a guess for the parameter vector θ, one computes the difference of model and data moments:

$$g_i(\theta) = m_i - \hat{m}_i(\theta)$$

$$g(\theta) = (g_1(\theta), ..., g_m(\theta)),$$

where $m = 13$ denotes the number of targeted moments. Then the SMM estimator $\hat{\theta}$ is given by:

$$\hat{\theta} = \arg\min_{\theta} \; g(\theta)' \, \mathcal{W} \, g(\theta),$$

where $\mathcal{W}$ is any positive definite matrix. Analogous to the GMM estimation, the optimal weighting matrix is given by $\mathcal{W} = S^{-1}$, where S is the variance-covariance matrix estimated from real data.

Standard Errors. In order to compute standard errors for the estimates, I generate N repetitions of model moments from T length simulated data The variance-covariance matrix of estimator $\hat{\theta}$ is

$$V = \left(1 + \frac{1}{N}\right) \left[\left(\frac{\partial g(\hat{\theta})}{\partial \theta} \right)' W \left(\frac{\partial g(\hat{\theta})}{\partial \theta} \right) \right]^{-1}, \tag{15}$$

where $\dfrac{\partial g(\hat{\theta})}{\partial \theta}$ is the Jacobian matrix, which can be obtained using standard numerical techniques. Finally, to get standard errors,

$$std(\hat{\theta}_i) = [V]_{ii}.$$

The intuition behind equation (15) is that if moment condition is very sensitive to changes in θ, then standard errors will be low.

Regarding the data targets, a discussion of the identification strategy is in order. It is well understood that changes to any of the parameters considered in estimation affect the model outcomes for all of the targeted moments. However, some moments are more responsive to certain parameters. Heuristically a moment target is informative about an unknown parameter if that target is sensitive to changes in the parameter value. Before starting the discussion, let me cathegorize the parameters estimated inside the model as

either real parameters or financial parameters.

Real parameters. To pin down the returns to scale and the depreciation rate (α, δ) I match the variance of investment and the average annual investment rate defined as the ratio of total investment to total capital stock.

The autocorrelation of investment and the autocorrelation of EBIT help identify ρ, the persistence of the firm profit shock. The standard deviation of investment and the volatility of cash flow provide instead information about σ_ε, the dispersion of the firm profit shock. As an additional restriction to identify the shocks parameters $(\rho, \sigma_\varepsilon)$, I run a panel autoregression of operating income zk^α on lagged operating income, and match the autoregressive coefficient (i.e. the coefficient on lagged operating income) and the standard deviation of the regression residual (see Hennessy and Whited 2007).

To pin down the parameters of the capital adjustment cost function (ψ_0, F), I match the frequency of investment inaction and the average cash/assets. Furthermore, the quadratic adjustment cost affects also the volatility of investment rate and the autocorrelation: a higher ψ_0 makes investment less volatile and more autocorrelated. Higher ψ_0 also implies a lower mean equity issuances and lower cash holdings. The opposite happens with F. A higher F makes the firm invest less frequently. The episodes of investment become though more pronounced so the firm is more likely to tap external equity finance, and in turn increases the precautionary motive for holding cash.

Financial parameters. In order to pin down the three parameters of the external finance cost function $(\lambda_0, \lambda_1, \lambda_2)$, I use targets that are informative about the financial choices of the firm. In particular I ask the model to match the mean of equity issuances[24], the standard deviation of equity issuances and the frequency of equity issuances. This latter moment is very infomative about the size of the fixed cost of raising new equity, λ_0. The variance of equity issuances is instead very sensitive to changes in λ_2. See also Appendix D for more details on the identification.

4.3 Estimation Results

Turning now to the results from the estimation exercise, Table 5 presents the point estimates and the standard errors, whereas Table 6 shows the goodness of fit of the model comparing the real targets to the model-simulated targets.

The estimated parameters together with their standard errors are presented in Table 5. The point estimates of the profit function curvature, the depreciation rate and the

[24] All moments are defined as ratio to total assets.

serial correlation of the profit shock are qualitatively in line with Hennessy and Whited (2007) and DeAngelo et al. (2011). The estimates of the adjustment costs of capital are also comparable to those in the literature using Compustat data, even though they differ significantly from Cooper and Haltiwanger (2006). With respect to the latter study my estimates for the fixed cost are lower and the estimate for the quadratic cost are higher. Two reasons can explain such discrepancy: First, the data used here are based on publicly traded firms whose investment behaviour is typically smoother than in plant-level observations from the Longitudinal Research Database used in Cooper and Haltiwanger (2006). Second, the adjustment costs parameters in my model influence greatly also the ratio of cash to assets, as it was shown in Figure 10.

Turning to the financial side of the model economy, the parameters λ_0 and λ_1 are both statistically significant at the 5% level. Although λ_2 is positive, it is statistically insignificant. Overall the estimates of the external equity cost function are somewhat smaller than Hennessy and Whited (2007). However, given that the effect of increasing uncertainty on cash holdings is larger the higher the external finance costs, my result is on the conservative side.

As shown in Table 6 my model fits the data reasonably well[25]. Most simulated moments in Table 6 match the corresponding data moments well, with the relevant exceptions of (i) the average size of equity issuances and (ii) the standard deviation of investment. Regarding (i), the model significantly underpredicts mean equity issuances: lowering the cost of external finance would of course raise equity issuances but at the same time would worse the model fit along other dimensions. Finally, on (ii), adding some extra friction to the capital adjustment process would help match the volatility of the investment rate more closely.

Overall the model is able to reproduce key features of the data. This result strengthens the reliability of using the estimated models to examine the relationship between firm level uncertainty and firm's cash holdings.

4.4 Out-of-Sample Fit: Regressions of Cash on Firm-level volatility

In order to further validate the model fit along non targeted statistics, I use the model to simulate a panel of firms imitating my Compustat sample. Then I run the same regressions both in the real Compustat data and in the simulated data from the model.

[25]The reader should bear in mind that the model is highly nonlinear and overidentified.

Table 5: Estimated parameters: results of SMM procedure

Parameter	**Description**	**Value**	**Standard Error**
Real parameters			
α	Profit function curvature	0.849	0.273
δ	Depreciation rate	0.133	0.033
σ_ε	Shock volatility	0.300	0.085
ρ	Shock autocorrelation	0.758	0.210
ψ_0	Quadratic adjustment cost	0.236	0.107
F	Fixed adjustment cost	0.0012	0.00048
Financial parameters			
λ_0	Fixed cost of external finance	0.589	0.302
λ_1	Linear cost of external finance	0.053	0.024
λ_2	Quadratic cost of external finance	0.0004	0.0005

Table 6: Model Fit: Comparing Real Moments vs Simulated Moments

Target	**Data (1980-1985)**	**Model**
Aggregate I/K	0.1334	0.1334
Average Cash	0.1037	0.0996
Average Equity Iss.	0.0371	0.0150
Average Q	1.3764	2.1609
Average EBIT	0.0978	0.0813
StDev Cash	0.1259	0.1284
StDev Invest	0.1155	0.1763
StDev Equity Iss.	0.1082	0.0979
StDev EBIT	0.1044	0.0912
StDev Cash Flow	0.0784	0.0794
Average (invest = 0)	0.0519	0.0659
Average (eqiss > 0)	0.2509	0.2438
Autocorr EBIT	0.6932	0.6905

Notes: The table compares real data moments and the corresponding moments obtained from simulations. Average Q: average Tobin's Q, Average (invest = 0): average frequency of investment inaction, Average (eqiss > 0): average share of firms issuing equity, EBIT: earnings before interests and taxes. For more details on variables definition, see Appendix C.

In the data I find a positive effect of firm level risk on cash holdings, which is stronger for financially constrained firms and for firms with more lumpy investment, as shown in the first column of Table 7. The model produces coefficients which all have the same sign as in the data, and that's reassuring about the model validation. However the coefficients are in larger magnitude in the model regressions than in the data regressions. Why? The model is a parsimonious description of the real world and the regressors in the model naturally contain more information (there is less noise than in the data).

Table 7: Comparing regressions on real data and model data

Cash	**Data**	**Model**
FLV	0.085***	0.267***
FLV × *Fin*	0.033**	0.156***
FLV × *Lumpy*	0.027**	0.099***
Controls	Yes	Yes
R^2	0.267	0.314

*** p<0.01, ** p<0.05, * p<0.1

Notes: FLV denotes firm level risk. *Fin* is financial constraint dummy (dividend payer status). *Lumpy* is real investment friction dummy (investment inaction). Firm level controls include Tobin Q, firm size, cash flow, capital expenditures, dividend.

4.5 Counterfactual Experiments: explaining the corporate cash increase

I can now use the model economy estimated on the time span 1980-1984 for counterfactual analysis. I perform a set of counterfactual experiments in order to quantify the relative importance of increasing firm-level volatility on the size of rising cash holdings over the 1980-2010 period. Table 8 summarizes the main quantitative results. In the main experiment reported in panel (a) of Table 8 I increase the firm shock volatility σ_ε from the estimated value of 0.3 to 0.45 in order to generate the exact same increase in cash flow voaltility that is observed in the data, from the 1980-1984 to the 2006-2010 time span. For the sake of clarity all other model parameters are kept at their initial values (estimated on the 1980-1984 data): in this way the effect of rising uncertainty is completely isolated from other shocks that may have hit the U.S. economy over the same time horizon. As a further robustness check I also perfom another exercise in which I recalibrate also the other parameters to match the new steady state and the results do not change dramatically[26].

[26]Results are available upon request.

First consider the baseline economy with both real and financial frictions (non-convex adjustment costs of capital plus costly equity financing) in panel (a) of table 8. The first row shows that, as firm-level uncertainty increases from 0.3 to 0.45, the cash/assets ratio in the model increases from 9% to 19%. This result shows that an increase in the firm precautionary demand for cash can explain around 87% of the increase in cash observed in the data. In fact, over the 1980-2010 period, the cash ratio in the data increased by 12 percentage points, from 9% to 21%, suggesting that the estimated model with real and financial frictions can account for the main share of the increase in cash holdings. As a validation of these results, Table 9 shows how all the other data moments change after the shock standard deviation is increased from 0.3 to 0.45 to match the increase in cash flow volatility in the data. The increase in cash flow volatility is matched exactly by construction (as explained above) but also most of the other moments change in the correct direction. This provides further reassurance about the validity of the model.

The empirical analysis in Section 2 pointed out the significant heterogeneity between industries in the rise in cash holdings: sectors that vary a great deal in terms of lumpiness and investment irreversibility may have a different reaction to the economy-wide increase in uncertainty. In order to explore the interplay between the increase in CFV and investment lumpiness I perform a second counterfactual experiment, presented in panel (b) of Table 8, In this second experiment I again increase the dispersion parameter in the model to generate the same increase in uncertainty that is observed in the data, but this time I completely shut down the non-convex adjustment costs channel, by setting $F = 0$. It is clear that financial frictions (or, to be more precise, the precautionary motive coming from financial frictions) alone are not sufficient to match the magnitude of the increase in corporate cash holdings observed in the data: as firm level risk increases from 0.15 to 0.45, the cash-to-assets ratio increases by only 5 percentage points. This result shows that the interaction between financial and real frictions is key to generate a sufficient motive for cash hoarding.

Table 8: Firm-Level Risk and Cash Holdings: Results from the Model

	Fin.Frictions + Fixed costs			**Fin.Frictions w/out Fixed costs**		
Firm Volatility σ_ε	0.30	0.45		0.30	0.45	
Cash/Assets (US data)	0.09	0.22	+133%	0.09	0.22	+133%
Cash/Assets (model)	0.09	0.19	+116%	0.09	0.14	+55%
Perc. explained by model			87.22%			48.67%

Notes: The table shows the results of two counterfactual experiments. The left panel is based on the baseline model with financial frictions and real investment frictions (in the form of fixed costs of adjusting capital). The right panel is based on an alternative scenario without real investment frictions.

Table 9: Model validation.

Target	**Model experiment**	**Data 2010**
Aggregate I/K	0.105	0.096
Average Cash	0.19	0.22
Average Equity Iss.	0.062	0.06
Average Q	2.371	1.84
Average EBIT	0.025	0.017
StDev Cash	0.340	0.238
StDev Invest	0.323	0.101
StDev Equity Iss.	0.132	0.174
StDev EBIT	0.182	0.219
StDev Cash Flow	**0.258**	**0.258**
Average (invest=0)	0.073	0.10
Average (eqiss¿0)	0.385	0.296
Autocorr EBIT	0.791	0.773

Notes: The table shows how all model moments change after shock volatility σ_ϵ is increased to match the cash flow volatility increase in the data (**StDev Cash Flow** is matched by construction).

5 Conclusions

In this paper I have presented new evidence which supports the hypothesis that the rise in firm levell volatility can explain the secular increase in US corporate cash holdings over the last four decades. The empirical analysis shows that firm level uncertainty is a key empirical determinant of firm level cash holdings. In addition, the available evidence suggests that both financial and real frictions contribute substantially to explain why firm level uncertainty matters so much. Regarding real frictions, I document a novel stylized fact: the impact of firm level uncertainty on cash holdings is stronger for firms that belong to sectors with greater investment lumpiness (measured by the industry frequency of investment inaction and by the industry investment spikes indicators).

Next, I build a structural dynamic model where uncertainty induces a precautionary behaviour: firms retain cash to avoid raising costly external finance when positive investment opportunities arrive. I show quantitatively that such precautionary effect depends strongly on the presence of fixed capital adjustment costs. Indeed in the baseline simulation, when I increase firm volatility to match the corresponding increase in firm volatility in the US data, the model is able to generate a substantial increase in cash-to-asset ratios only when non-convexities in adjustment costs are present.

Finally, this paper also sheds some light on the recent policy debate about corporate firms' cash holdings and investment. Policy makers and commentators[27] have expressed concerns as to why firms are building large stockpiles of cash instead of investing it or distributing it back to shareholders. My analysis shows that cash is a useful tool to insure against increasing uncertainty: having a buffer of cash enables firms to invest when they need it. In this sense the increasing hoarding of cash and liquid assets by U.S. corporations is an efficient response to an economic environment that has become more risky over time.

[27] See for example "What is business waiting for?" by Joe Nocera, *The New York Times*, August 15, 2011, accessible here.

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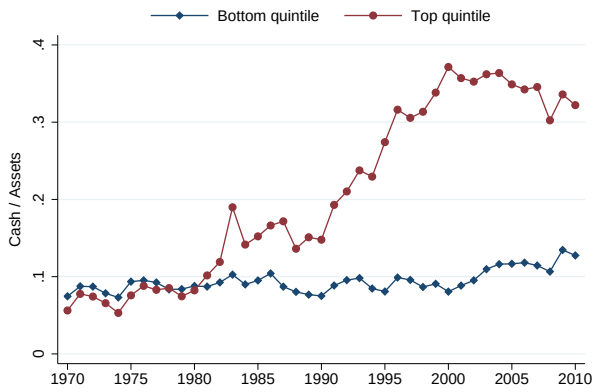
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Appendix

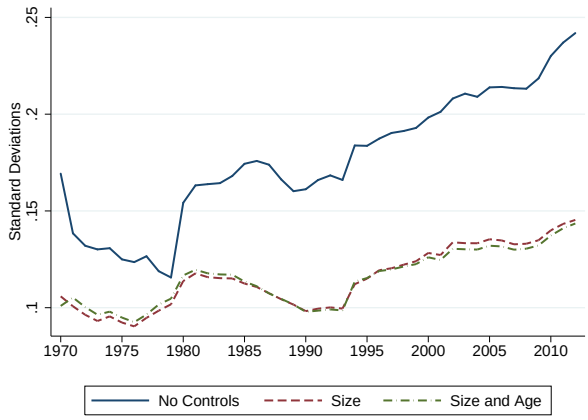
A Additional figures

Figure 11: Cash-to-assets ratio and Investment Lumpiness



Notes: Firms sorted by frequency of Investment Inaction. Data source: U.S. Compustat Annual, excluding financial companies and utilities.

Figure 12: Firm-level Risk, controlling for size and age



Notes: Firm level risk, 10 year rolling window. Robustness check including (i) `log(size)` and (ii) `log(size)` and `log(age)` to control for compositional bias. Data source: U.S. Compustat Annual, excluding financial companies and utilities.

B Optimal Cash Policy

In this appendix I present a detailed derivation of the analytical results shown in Section 3.3. For convenience I define:

$$e\left(k,k',p,p',z\right)\equiv\left(1-\tau_c\right)\pi\left(k,z\right)+\tau_c\delta k-\left[k'-\left(1-\delta\right)k\right]-\psi\left(k',k\right)-\left[p'-\left(1+r\left(1-\tau_c\right)\right)p\right]$$

and

$$e\left(k',k^{''},p',p'',z'\right)\equiv\left(1-\tau_c\right)\pi\left(k',z'\right)+\tau_c\delta k'-\left[k''-\left(1-\delta\right)k'\right]-\psi\left(k'',k'\right)-\left[p''-\left(1+r\left(1-\tau_c\right)\right)p'\right].$$

Case $\tau_d=0$

If external finance is costless (i.e. $\lambda_1=\lambda_2=0$), it is easy to see that the optimality condition for p' holds a strict inequality. In this case we have:

$$MB=\frac{1+r\left(1-\tau_c\right)}{1+r}<1=MC,$$

hence $p'=p=0$.

In the more interesting case where $\lambda_1>0$ and $\lambda_2>0$, the first order condition with respect to p' is

$$1+\left(\lambda_1-\lambda_2 e\right)\mathbb{1}_{\{e<0\}}\geq\frac{1}{1+r}E_{z'|z}\left[V_p\left(k',p',z'\right)\right],$$

with equality if $p'>0$. The envelope condition is

$$V_p\left(k,p,z\right)=\left(1+r\left(1-\tau_c\right)\right)\left[1+\left(\lambda_1-\lambda_2 e\right)\mathbb{1}_{\{e<0\}}\right].$$

Updating the envelope condition and plugging it into the first order condition delivers:

$$1+\left(\lambda_1-\lambda_2 e\right)\mathbb{1}_{\{e<0\}}\geq\left(\frac{1+r\left(1-\tau_c\right)}{1+r}\right)E_{z'|z}\left[1+\left(\lambda_1-\lambda_2 e'\right)\mathbb{1}_{\{e'<0\}}\right],$$

with equality if $p'>0$. This analysis points out that the marginal values are:

$$MC=\begin{cases}1 & \text{if } e\geq 0\\ 1+\lambda_1-\lambda_2 e>1 & \text{if } e<0\end{cases}$$

and

$$MB=\begin{cases}\frac{1+r(1-\tau_c)}{1+r}, & \text{if } e'\geq 0\\ \left(\frac{1+r(1-\tau_c)}{1+r}\right)E_{z'|z}\left[1+\left(\lambda_1-\lambda_2 e'\right)\right] & \text{if } e'<0.\end{cases}$$

Lemma B.1. *MC is weakly increasing in* p'.

Proof. If p' increases, the payoff e decreases (and becomes negative). In particular there exists a threshold $\widetilde{p}'$ such that $e\left(k,k',p,p',z\right)\geq 0$ if $p'\leq\widetilde{p}'$ and $e\left(k,k',p,p',z\right)<0$ if $p'>\widetilde{p}'$. Indeed

$$\begin{aligned}e\geq 0 &\implies \left(1-\tau_c\right)\pi\left(k,z\right)+\tau_c\delta k-I-\psi\left(k',k\right)-\left[p'-\left(1+r\left(1-\tau_c\right)\right)p\right]\geq 0\\ &\implies p'\leq\left(1-\tau_c\right)\pi\left(k,z\right)+\tau_c\delta k-I-\psi\left(k',k\right)+\left(1+r\left(1-\tau_c\right)\right)p\equiv\widetilde{p}'.\end{aligned}$$

□

Lemma B.2. *MB is decreasing in p'.*

Proof. A higher value for future cash implies that the probability of $\{e' < 0\}$ is lower, hence MB goes down. A higher z implies higher investment, which leads to a lower $\widetilde{p}'$. This increases the MC of cash, hence the optimal choice for p' decreases. □

Case $\tau_d > 0$

In this case the FOC wrt p' is

$$1 - \tau_d \mathbb{1}_{\{e \geq 0\}} + (\lambda_1 - \lambda_2 e)\, \mathbb{1}_{\{e<0\}} \geq \frac{1}{1+r} E_{z'|z} \left[V_p \left(k', p', z' \right) \right],$$

with equality if $p' > 0$. The envelope condition is

$$V_p \left(k, p, z \right) = \left(1 + r \left(1 - \tau_c \right) \right) \left[1 - \tau_d \mathbb{1}_{\{e \geq 0\}} + (\lambda_1 - \lambda_2 e)\, \mathbb{1}_{\{e<0\}} \right].$$

Updating the envelope condition and plugging it into the FOC gives:

$$1 - \tau_d \mathbb{1}_{\{e \geq 0\}} + (\lambda_1 - \lambda_2 e)\, \mathbb{1}_{\{e<0\}} \geq \left(\frac{1 + r \left(1 - \tau_c \right)}{1+r} \right) E_{z'|z} \left[1 - \tau_d \mathbb{1}_{\{e' \geq 0\}} + (\lambda_1 - \lambda_2 e')\, \mathbb{1}_{\{e'<0\}} \right],$$

with equality if $p' > 0$. Hence the marginal values are:

$$MC = \begin{cases} 1 - \tau_d < 1 & \text{if } e \geq 0 \\ 1 + \lambda_1 - \lambda_2 e > 1 & \text{if } e < 0 \end{cases}$$

and

$$MB = \begin{cases} \left(\frac{1+r(1-\tau_c)}{1+r} \right) (1 - \tau_d) < 1 - \tau_d, & \text{if } e' \geq 0 \\ \left(\frac{1+r(1-\tau_c)}{1+r} \right) E_{z'|z} \left[1 + (\lambda_1 - \lambda_2 e') \right] & \text{if } e' < 0. \end{cases}$$

C Data

The data used in the paper are from Compustat annual database for the period 1980-2010. I exclude firm-year observations for which total assets, sales or capital are either negative or missing. I also drop observations with total assets less than 10 million USD. Finally I delete firms belonging to SIC codes 6000-6999, 4900-4999 or greater than 9000 since my model is ill-suited to study regulated, financial firms or utilities. I trim the top and bottom 2% of the variables following Hennessy and Whited (2007). After the data cleaning criteria are applied, there are a total of 133803 firm-year observations with 14256 unique firms. A detailed description of variables definition follows.

Total Assets: Total assets are defined as Compustat variable `AT` (data6). In the model, total assets are obtained by summing physical assets k and cash p.

Cash Holdings: Cash holdings are computed as cash and marketable securities (data item `CHE`) divided by total assets (data item `AT`). In the model cash holdings are:

$$\frac{p}{k+p},$$

where total assets = capital + cash holdings. Cash ratio is computed in the data as:

$$\texttt{Cash}_t = \frac{1}{N}\sum_i \frac{\texttt{Cash}_{it}}{\texttt{AT}_{it}}$$

Figure 2b shows instead the aggregate cash to aggregate asset ratio:

$$\texttt{Agg.Cash}_t = \frac{\sum_i \texttt{Cash}_{it}}{\sum_i \texttt{AT}_{it}}.$$

The latter measure is just a weighted version of the former:

$$\begin{aligned}\texttt{Weighted.CashRatio}_t &= \sum_i \frac{\texttt{Cash}_{it}}{\texttt{AT}_{it}} \cdot \frac{\texttt{AT}_{it}}{\sum_j \texttt{AT}_{jt}} \\ &= \sum_i \frac{\texttt{Cash}_{it}}{\sum_j \texttt{AT}_{jt}} = \texttt{Agg.Cash}_t.\end{aligned}$$

Net leverage: Net leverage at the firm level is computed as the ratio of long-term debt (item `DLTT`) plus debt in current liabilities (item `DLC`) minus cash and liquid financial assets (item `CHE`) to total book assets (item `AT`).

Investment: It is computed as the ratio between capital expenditures (data item `CAPX`) and total assets (data item `AT`). Investment in the model is defined as:

$$ik = \frac{I}{k+p} = \frac{k' - (1-\delta)\,k}{k+p}.$$

Sometimes it is better to define the investment rate as I/k, excluding the cash stock p from the denominator. Average investment can be computed in two ways (both in the data and in the model). Average cross-sectional investment:

$$E\left(ik\right) = \frac{1}{T}\sum_t \frac{1}{N}\sum_i \frac{I_{it}}{k_{it}} = \frac{1}{TN}\sum_{i,t}\frac{I_{it}}{k_{it}}$$

or average investment ratio:

$$E\left(IK\right) = \frac{1}{T}\sum_t \left[\frac{I_t}{K_t}\right] = \frac{1}{T}\sum_t \left[\frac{\sum_i I_{it}}{\sum_t k_{it}}\right]$$

The investment standard deviation is computed as follows:

$$InvSTDEV = \frac{1}{TN}\sum_{i,t}\left(\frac{I_{it}}{k_{it}}\right)^2 - \left[E\left(\frac{I_{it}}{k_{it}}\right)\right]^2,$$

hence it can be interpreted as the standard deviation of the pooled data.

EBITDA: Earnings before interests taxes and depreciation are defined as the data item `OIBDP`. The Compustat variable `OIBDP` is obtained as Sales (`SALE`, data12) - Cost of Goods Sold (`COGS`, data41) - Selling, General and Administrative Expense (`XSGA`, data189). See Table 10 for more details. The model counterpart of `OIBDP` (and, hence, *EBITDA*) is simply:

$$EBITDA = \frac{zk^{\alpha}}{k+p}.$$

EBIT: Earnings before interest expenses and taxes are defined as operating income minus depreciation. In the model it is as follows:

$$EBIT = \frac{zk^{\alpha} - \delta k}{k+p}.$$

Cash Flow: It is defined as operating income (sales minus operating costs) after interest and taxes but before depreciation. In Compustat we have Cash Flow = Operating Income (`OIBDP`, data13) - Interest Expense (`XINT`, data15) - Income Taxes (`TXT`, data16). By construction this is equal to Income before extra items (`IB`, data18) + Depreciation (`DP`, data14)

Dividend: Dividends are measured as commond dividends (data item `DVC`).

Distributions. Total distributions are the sum of item 19 (`DVP`), item 21 (`DVC`), item 115 (`PRSTKC`).

Equity issuances. Equity issuances are defined as item 108 (`SSTK`) or net as `SSTK-PRSTKC`.

Frequency of investment inaction. Investment inaction is defined at the sectoral level following Cooper and Haltiwanger (2006) as the number of firm-year observations such that |CAPX/AT| $< 1\%$ divided by the total number of observations in that sector.

Investment Spikes. It is defined at the sectoral level following Cooper and Haltiwanger (2006) as the number of firm-year observations such that |CAPX/AT| $> 20\%$ divided by the total number of observations in that sector.

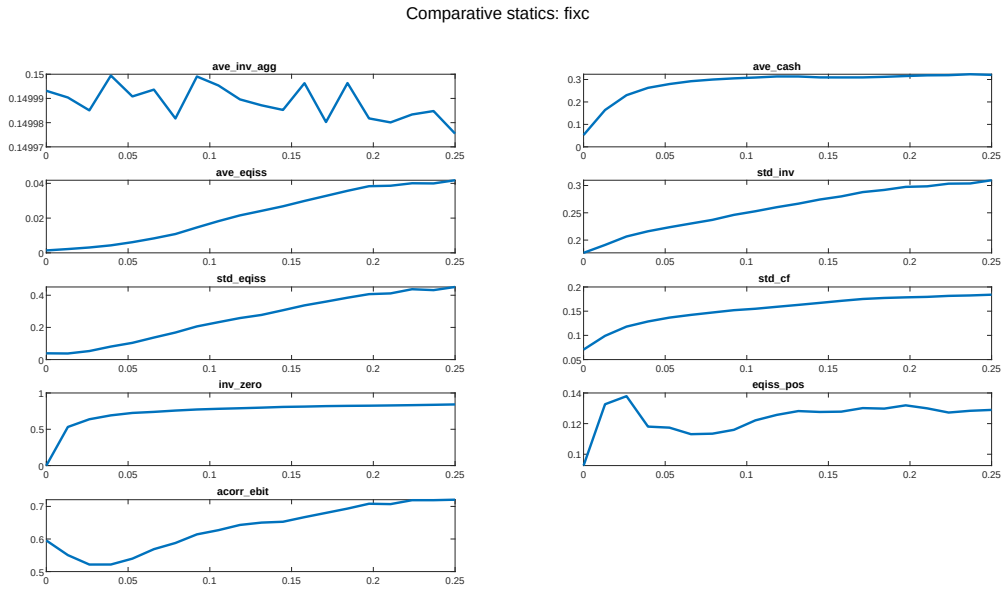
Table 10: Firm's balance sheet in Compustat

+**SALE**	Revenue**s**
−COGS	cost of goods sold
−XSGA	Selling, General, and Admin Expense
= **OIBDP**	operating income before depreciation (*EBITDA*)
−DP	depreciation
= **OIADP**	operating income after depreciation (*EBIT*)
−XINT	interest expenses
−TXT	taxes paid
=**IB**	income before extraord items

D Identification

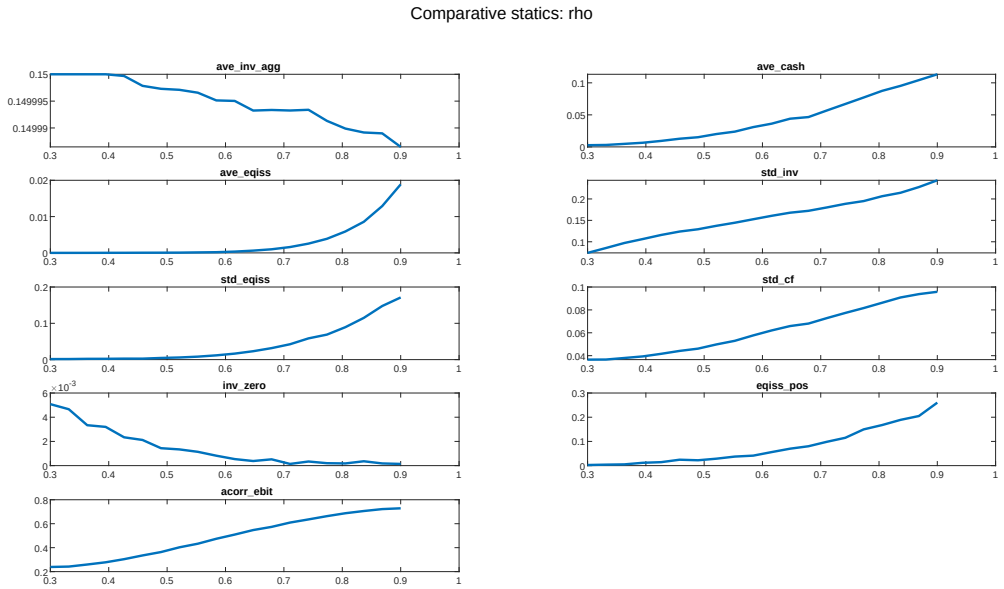
In this section I report some plots that help understand how model parameters are identified. Each panel shows how a selected number of moments change as the parameter of interest changes around its estimated value. The idea underlying such analysis is that a parameter is well identified whenever it has a monotonic effect on the chosen targets.

Figure 13: Sources of Identification: F



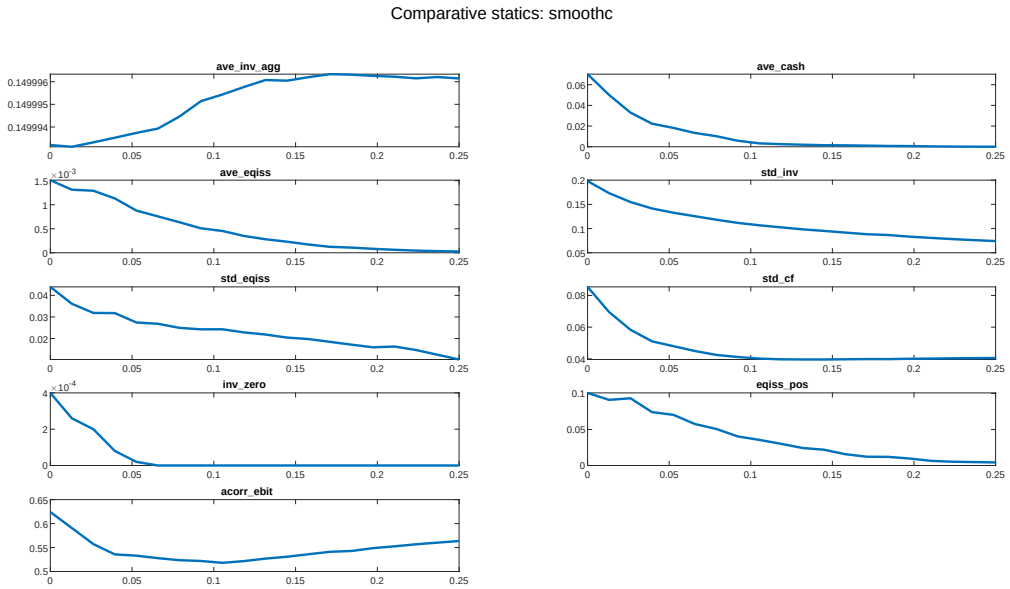
Notes: The figure shows how the fixed cost of capital adjustment F affects a number of selected moments from the model. These are the averages of investment rate, cash balances, equity issuances, (`ave_inv_agg`, `ave_cash` and `ave_eqiss`, respectively) the standard deviations of investment, cash flows and equity issuances (`std_inv`, `std_eqiss` and `std_cf`), the share of firms with investment inaction, with positive equity issuances (`inv_zero`, `eqiss_pos`), and the autocorrelation of EBIT (`acorr_ebit`).

Figure 14: Sources of Identification: ρ



Notes: The figure shows how the persistence of idiosyncratic shocks ρ affects a number of selected moments from the model. For a description of the model moments, see notes to Figure 13.

Figure 15: Sources of Identification: ψ_0



Notes: The figure shows how the quadratic cost of capital adjustment ψ_0 affects a number of selected moments from the model. For a description of the model moments, see notes to Figure 13.